MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Lesson 23: Cardinal Numbers: Relations Between Sets (Summarized Notes)

Cardinality is used to compare the sizes of sets.

Let A and B be two sets.

We write $|A| \leq |B|$ if there exists a one-to-one (injective) function $f : A \to B$.

An injective function means every element of A maps to a different element of B.

The relation $|A| \leq |B|$ is well-defined.

It does not depend on the particular sets chosen.

It depends only on their cardinal numbers.

Example 1

Let A be a proper subset of a finite set B.

Since every element of A is already in B, the inclusion function is one-to-one.

Therefore,

$|A| < |B|$

A proper subset of a finite set always has fewer elements than the original set.

Example 2

Let **n** be a finite cardinal number.

If A has n elements and B has more than n elements, then

$|A| = n \leq |B|$

A finite set cannot have more elements than a larger finite set.

Example 3

Consider

P = {1, 2, 3, ...}

I = [0,1]

Define the function

f(n) = 1/n

Different positive integers give different values.

Therefore, the function is one-to-one (injective).

Hence,

|P| ≤ |I|

However,

|P| ≠ |I|

The interval [0,1] contains infinitely more real numbers than the set of positive integers.

Therefore,

|P| < |I|

This shows that not all infinite sets have the same size.

Example 4

Let A be any infinite set.

Every infinite set contains a denumerable (countably infinite) subset.

Therefore,

|P| ≤ |A|

Every infinite set has at least as many elements as the set of positive integers.

Important Exam Points

Cardinality compares the sizes of sets.

|A| ≤ |B| means there exists an injective (one-to-one) function from A to B.

Injective function means different inputs have different outputs.

The relation **|A| ≤ |B|** is independent of the actual sets and depends only on their cardinalities.

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A proper subset of a finite set has strictly smaller cardinality.

The function **f(n) = 1/n** from P to I is injective.

The interval **[0,1]** is uncountable.

The set of positive integers **P** is countably infinite (denumerable).

|P| < **|I|** because the real numbers in [0,1] are uncountable.

Every infinite set contains a countably infinite subset.

For every infinite set A,

|P| ≤ |A|

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5 Very Important MCQs

1. The notation |A| ≤ |B| means that there exists a

A) Onto function

B) One-to-one function ✅

C) Constant function

D) Identity function

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2. The function f(n) = 1/n from P to I is

A) Many-to-one

B) Onto

C) One-to-one ✅

D) Constant

---

3. The set of positive integers P is

A) Finite

B) Empty

C) Denumerable ✅

D) Uncountable

---

4. Which relation is correct?

A) |I| < |P|

B) |P| = |I|

C) |P| < |I| ✅

D) |P| > |I|

---

5. Every infinite set contains a

A) Finite subset only

B) Denumerable subset ✅

C) Empty subset only

D) Singleton subset only

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10 Very Important Mathematical Examples

Example 1

A = {1,2}

B = {1,2,3}

Since A is a proper subset of B,

|A| < |B|

Example 2

A = {a,b,c}

B = {1,2,3,4}

Define

f(a)=1

f(b)=2

f(c)=3

All images are different.

Therefore, f is injective.

Hence,

|A| ≤ |B|

Example 3

Find f(5) if

f(n)=1/n

Solution

f(5)=1/5

Example 4

Find f(8).

Solution

f(8)=1/8

Example 5

Show that f(2)=f(4)?

f(2)=1/2

f(4)=1/4

Since

1/2 ≠ 1/4

The function is one-to-one.

Example 6

Suppose

f(3)=1/3

f(6)=1/6

Since

1/3 ≠ 1/6

Different inputs give different outputs.

Hence, f is injective.

Example 7

A has 5 elements.

B has 8 elements.

Since 5 ≤ 8,

|A| ≤ |B|

Example 8

P = {1,2,3,...}

I = [0,1]

Using

$f(n)=1/n$

There exists an injective function.

Therefore,

$|P| \leq |I|$

Example 9

Since the interval [0,1] is uncountable,

$|P| \neq |I|$

Hence,

$|P| < |I|$

Example 10

Let A be an infinite set.

Every infinite set contains a denumerable subset.

Therefore,

$|P| \leq |A|$

LESSON 24: Cantor's Theorem (Summarized Notes)

Cantor's Theorem states that for every set A, the power set P(A) always contains more elements than A itself.

The power set P(A) is the set of all possible subsets of A.

Every subset of A is an element of P(A).

If a set A contains **n** elements, then the total number of subsets is **2^n**.

Therefore, the cardinality of the power set is **$|P(A)| = 2^n$**.

Since $\mathbf{2^n > n}$ for every positive integer n, the power set always has more elements than the original set.

Example:

If $\mathbf{A = \{1,2\}}$

Then

$\mathbf{P(A) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}}$

Number of elements in A = **2**

Number of elements in P(A) = **4**

Therefore,

$|\mathbf{P(A)}| > |\mathbf{A}|$

Cantor's Theorem is used to compare the cardinality of a set with the cardinality of its power set.

Important Exam Points

Cantor's Theorem states that the power set of every set has a greater cardinality than the set itself.

Power set means the set of all subsets of a given set.

Formula for the number of subsets:

$|\mathbf{P(A)}| = \mathbf{2^n}$

where **n = number of elements in A**.

The empty set ($\emptyset$) is always included in every power set.

The original set A is also always a member of its own power set.

For every finite set,

$|\mathbf{P(A)}| > |\mathbf{A}|$

Be able to construct the complete power set of a small set.

Remember that every additional element doubles the number of subsets.

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5 Very Important Statement-Based MCQs

1. Which of the following statements about Cantor's Theorem is TRUE?

A) Every set and its power set have the same number of elements.

B) The power set of a set always has fewer elements than the set.

C) The power set of every set has strictly greater cardinality than the set itself. ✅

D) A power set contains only proper subsets.

---

2. Which statement correctly defines the power set P(A)?

A) It is the set of all proper subsets of A.

B) It is the set of all supersets of A.

C) It is the set containing all subsets of A, including ∅ and A itself. ✅

D) It contains only singleton subsets.

---

3. Which of the following statements is CORRECT if a set contains n elements?

A) Number of subsets = n^2

B) Number of subsets = n + 2

C) Number of subsets = 2^n ✅

D) Number of subsets = n!

---

4. Which statement is TRUE for every power set?

A) It never contains the empty set.

B) It always contains the original set and the empty set. ✅

C) It contains only non-empty subsets.

D) It contains only subsets having one element.

---

5. According to Cantor's Theorem, if |A| = 3, then which statement is TRUE?

A) |P(A)| = 3

B) |P(A)| = 6

C) |P(A)| = 8, therefore |P(A)| > |A|. ✅

D) |P(A)| = 9

---

10 Very Important Mathematical Examples (Paper Pattern)

Example 1

Find the power set of

A = {1}

Solution

P(A) = {∅, {1}}

Number of subsets = $\mathbf{2^1 = 2}$

Example 2

Find the power set of

A = {1,2}

Solution

P(A) = {∅, {1}, {2}, {1,2}}

Number of subsets = $\mathbf{2^2 = 4}$

Example 3

Find the power set of

A = {a,b}

Solution

P(A) = {∅, {a}, {b}, {a,b}}

Total subsets = **4**

Example 4

A set contains **3** elements.

Find the number of subsets.

Solution

$|P(A)| = 2^3 = 8$

Example 5

A set contains **4** elements.

Find the number of subsets.

Solution

$|P(A)| = 2^4 = 16$

Example 6

A set contains **5** elements.

Find the cardinality of its power set.

Solution

$|P(A)| = 2^5 = 32$

Example 7

Verify Cantor's Theorem for

A = {a,b,c}

Solution

Number of elements in A = **3**

Number of subsets = $\mathbf{2^3 = 8}$

Since **8** > **3**

Therefore,

|**P(A)**| > |**A**|

Cantor's Theorem is verified.

Example 8

If

|**A**| = **6**

Find |**P(A)**|.

Solution

$\mathbf{|P(A)| = 2^6 = 64}$

Example 9

If

|**P(A)**| = **16**

Find the number of elements in A.

Solution

Since

$2^n = 16$

Therefore,

n = 4

Hence,

|A| = 4

Example 10

If

|P(A)| = 32

Determine **|A|**.

Solution

Since

$2^n = 32$

Therefore,

n = 5

Hence,

|A| = 5

LESSON 25: Addition and Multiplication of Cardinal Numbers (Summarized Notes)

Addition and multiplication of counting numbers can be explained using set theory.

The equation **2 + 3 = 5** means the union of two **disjoint sets**, one having 2 elements and the other having 3 elements, contains 5 elements.

This idea is generalized to define the addition of cardinal numbers.

Let **α** and **β** be cardinal numbers.

Let **A** and **B** be disjoint sets such that

$|A| = \alpha$

$|B| = \beta$

The sum of two cardinal numbers is defined as

$\alpha + \beta = |A \cup B|$

The sets **A** and **B** must be disjoint.

The definition of addition is well-defined.

If **m** and **n** are finite cardinal numbers, then cardinal addition is the same as ordinary addition in the counting numbers **N**.

Multiplication of cardinal numbers is defined using the Cartesian product.

Let

$|A| = \alpha$

$|B| = \beta$

Then

$\alpha\beta = |A \times B|$

The Cartesian product contains all ordered pairs **(a, b)** where **$a \in A$** and **$b \in B$**.

If **m** and **n** are finite cardinal numbers, then cardinal multiplication is the same as ordinary multiplication in **N**.

The Cartesian product **$N \times N$** is denumerable.

The Cartesian plane **$R \times R$** has the same cardinality as **R**.

Therefore,

$c \times c = c$

where **c** denotes the cardinality of the real numbers.

If **β** is an infinite cardinal number and **$\alpha \leq \beta$**, then

$\alpha + \beta = \beta$

and

$\alpha\beta = \beta$

Adding or multiplying a smaller cardinal number with an infinite cardinal number does not change the infinite cardinality.

Important Exam Points

Addition of cardinal numbers is defined by the union of two disjoint sets.

Formula for addition:

$\alpha + \beta = |A \cup B|$

The sets must be disjoint.

Multiplication of cardinal numbers is defined by the Cartesian product.

Formula for multiplication:

$\alpha\beta = |A \times B|$

For finite cardinal numbers, cardinal addition equals ordinary addition.

For finite cardinal numbers, cardinal multiplication equals ordinary multiplication.

The set $\mathbf{N} \times \mathbf{N}$ is denumerable.

The Cartesian plane satisfies

$\mathbf{R} \times \mathbf{R} = \mathbf{R}^2$

and

$\mathbf{c} \times \mathbf{c} = \mathbf{c}$

If β is infinite and $\alpha \leq \beta$, then

$\alpha + \beta = \beta$

and

$\alpha\beta = \beta$

These infinite cardinality properties are among the most important results of the lesson.

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5 Very Important Statement-Based MCQs

1. Which of the following statements correctly defines the addition of two cardinal numbers α and β?

A) **α + β = |A ∩ B|**

B) **α + β = |A × B|**

C) **α + β = |A ∪ B|**, where A and B are disjoint. ✅

D) **α + β = |A − B|**

---

2. Which statement about multiplication of cardinal numbers is TRUE?

A) It is defined using the union of sets.

B) It is defined using the Cartesian product of sets. ✅

C) It is defined using the intersection of sets.

D) It is defined using complements of sets.

---

3. Which of the following statements is CORRECT?

A) **N × N** is finite.

B) **N × N** is uncountable.

C) **N × N** is denumerable. ✅

D) **N × N** is empty.

---

4. Which statement correctly describes the Cartesian plane?

A) **c × c > c**

B) $c \times c = c$ ✅

C) $c \times c < c$

D) $c \times c = 2c$

---

5. Let β be an infinite cardinal number and $\alpha \le \beta$. Which statement is TRUE?

A) $\alpha + \beta = \alpha$

B) $\alpha\beta = \alpha$

C) $\alpha + \beta = \beta$ and $\alpha\beta = \beta$ ✅

D) $\alpha + \beta = \alpha\beta = \alpha^2$

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

$A = \{1,2,3\}$

$B = \{a,b,c,d\}$

where $A \cap B = \emptyset$.

Find $\alpha + \beta$.

Solution

$|A| = 3$

$|B| = 4$

Since the sets are disjoint,

$\alpha + \beta = |A \cup B|$

$= |\{1,2,3,a,b,c,d\}|$

= 7

Therefore,

3 + 4 = 7

Example 2

Let

A = {1,2,3}

B = {x,y}

Find **αβ** using the Cartesian product.

Solution

A × B

= {(1,x),(1,y),(2,x),(2,y),(3,x),(3,y)}

Number of ordered pairs

= **6**

Therefore,

3 × 2 = 6

Example 3

Suppose

|A| = 8

|B| = 5

and A and B are disjoint.

Without listing the elements, determine

|A ∪ B|

Solution

Using

$\alpha + \beta = |A \cup B|$

$= 8 + 5$

$= 13$

Example 4

Let

$|A| = 7$

$|B| = 9$

Determine

$|A \times B|$

Solution

Using

$\alpha\beta = |A \times B|$

$= 7 \times 9$

$= 63$

Example 5

Let $\alpha = 12$ and $\beta = \aleph_0$ (countably infinite), where $\alpha \leq \beta$.

Find $\alpha + \beta$ and $\alpha\beta$.

Solution

Since β is infinite,

$\alpha + \beta = \beta$

$\alpha\beta = \beta$

Therefore,

$12 + \aleph_0 = \aleph_0$

$12 \times \aleph_0 = \aleph_0$

Example 6

Show that **N × N** is countably infinite.

Solution

Given in the lesson,

N × N is denumerable.

Therefore,

$|\mathbf{N} \times \mathbf{N}| = |\mathbf{N}|$

Hence, the Cartesian product of two countably infinite sets remains countably infinite.

Example 7

The Cartesian plane is represented by **R × R**.

Determine its cardinality.

Solution

The lesson states

$\mathbf{R} \times \mathbf{R} = \mathbf{R}^2$

and

$|\mathbf{R}| = \mathbf{c}$

Therefore,

$|\mathbf{R} \times \mathbf{R}| = \mathbf{c} \times \mathbf{c} = \mathbf{c}$

Hence, the Cartesian plane has the same cardinality as the real numbers.

Example 8

Let $\boldsymbol{\alpha} = \boldsymbol{\aleph_0}$ and $\boldsymbol{\beta} = \mathbf{c}$, where $\boldsymbol{\alpha} \leq \boldsymbol{\beta}$.

Find $\boldsymbol{\alpha} + \boldsymbol{\beta}$.

Solution

Since $\boldsymbol{\beta}$ is infinite and $\boldsymbol{\alpha} \leq \boldsymbol{\beta}$,

$\boldsymbol{\alpha} + \boldsymbol{\beta} = \boldsymbol{\beta}$

Therefore,

$\boldsymbol{\aleph_0} + \mathbf{c} = \mathbf{c}$

Example 9

Let $\boldsymbol{\alpha} = \boldsymbol{\aleph_0}$ and $\boldsymbol{\beta} = \mathbf{c}$.

Find $\boldsymbol{\alpha\beta}$.

Solution

Since $\boldsymbol{\beta}$ is infinite and $\boldsymbol{\alpha} \leq \boldsymbol{\beta}$,

$\boldsymbol{\alpha\beta} = \boldsymbol{\beta}$

Therefore,

$\boldsymbol{\aleph_0} \times \mathbf{c} = \mathbf{c}$

Example 10

Let **A** be a finite set with **20** elements and **B** be a denumerable set.

Determine the cardinality of $\mathbf{A \cup B}$ and $\mathbf{A \times B}$.

Solution

Let

$|A| = 20$

$|B| = \aleph_0$

Since **$20 \leq \aleph_0$**,

$|A \cup B| = 20 + \aleph_0 = \aleph_0$

and

$|A \times B| = 20 \times \aleph_0 = \aleph_0$

Thus, both the union and the Cartesian product have **countably infinite cardinality**, illustrating the theorem that adding or multiplying a finite cardinal with an infinite cardinal does not change the infinite cardinality.

LESSON 26: Partial Order Relations (Summarized Notes)

A **partial order relation** is a relation defined on a set that satisfies three important properties.

Let **R** be a relation on a set **S**.

R is called a **partial order** if it satisfies the following properties.

Reflexive Property

For every element **$a \in S$**,

aRa

Every element is related to itself.

Antisymmetric Property

If

aRb and **bRa**

then

a = b

Two different elements cannot be related in both directions.

Transitive Property

If

aRb and **bRc**

then

aRc

The relation is preserved through intermediate elements.

A relation satisfying all three properties is called a **partial order** or **order relation**.

The set together with its partial order is called a **partially ordered set (Poset)**.

A partially ordered set is usually written as **(S, R)**.

If the ordering relation is already understood, we simply write **S**.

The most common partial ordering is the **usual order (≤)** on positive integers and real numbers.

For this reason, many partial orders are represented by the symbol ≤.

When using ≤, the three properties become:

a ≤ a (Reflexive)

If **a ≤ b** and **b ≤ a**, then **a = b** (Antisymmetric)

If **a ≤ b** and **b ≤ c**, then **a ≤ c** (Transitive)

The relation of **set inclusion (⊆)** is a partial ordering on any collection of sets.

The **divisibility relation (|)** on the positive integers is also a partial ordering.

If **a | b**, then there exists an integer **c** such that

ac = b

Examples include

2 | 4

3 | 12

7 | 21

The divisibility relation is **not** a partial ordering on the set of integers **Z** because it is **not antisymmetric**.

For example,

2 | -2

and

-2 | 2

but

2 ≠ -2

Hence, the antisymmetric property fails.

Important Exam Points

A partial order must satisfy **three properties only**.

Reflexive means **every element is related to itself**.

Antisymmetric means **if aRb and bRa, then a = b**.

Transitive means **if aRb and bRc, then aRc**.

A set together with a partial order is called a **Partially Ordered Set (Poset)**.

Notation for an ordered set:

(S, R)

The usual ordering ≤ is a partial order.

The inclusion relation ⊆ is a partial order on collections of sets.

The divisibility relation **(|)** is a partial order on **positive integers (P)**.

The divisibility relation is **not** a partial order on **integers (Z)** because it violates the antisymmetric property.

Always check **all three properties** before concluding that a relation is a partial order.

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5 Very Important Statement-Based MCQs

1. Which of the following statements is TRUE about a partial order relation?

A) It must satisfy only reflexivity.

B) It must satisfy reflexivity, antisymmetry, and transitivity. ✅

C) It must satisfy symmetry and transitivity only.

D) It must satisfy reflexivity and symmetry only.

---

2. Which statement correctly describes the antisymmetric property?

A) If **aRb**, then **bRa**.

B) If **aRb** and **bRa**, then **a = b**. ✅

C) Every element is related to itself.

D) If **aRb** and **bRc**, then **aRc**.

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3. Which of the following relations is a partial order on a collection of sets?

A) Equality

B) Set inclusion (⊆) ✅

C) Set union

D) Set difference

---

4. Which statement is TRUE regarding the divisibility relation (|)?

A) It is a partial order on all integers Z.

B) It is a partial order on positive integers P. ✅

C) It is symmetric on positive integers.

D) It is not reflexive.

---

5. Why is the divisibility relation not a partial order on the set of integers Z?

A) It is not reflexive.

B) It is not transitive.

C) It violates the antisymmetric property. ✅

D) It violates the reflexive property.

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10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether the relation ≤ on

S = {2,4,6,8}

is a partial order.

Solution

Reflexive:

$2 \leq 2$, $4 \leq 4$, $6 \leq 6$, $8 \leq 8$ ✓

Antisymmetric:

If $a \leq b$ and $b \leq a$, then $a = b$ ✓

Transitive:

If $2 \leq 4$ and $4 \leq 8$, then $2 \leq 8$ ✓

Therefore,

≤ is a partial order.

Example 2

Let

A = {1,2}

B = {1,2,3}

C = {1,2,3,4}

Verify transitivity for **⊆**.

Solution

$A \subseteq B$

$B \subseteq C$

Therefore,

$A \subseteq C$

Hence,

⊆ satisfies the transitive property.

Example 3

Verify that divisibility is a partial order on

P = {1,2,4,8}

Solution

1 | 1, 2 | 2, 4 | 4, 8 | 8 ✓

If 2 | 4 and 4 | 2, this never occurs unless both numbers are equal ✓

If 2 | 4 and 4 | 8,

then

2 | 8 ✓

Hence,

Divisibility is a partial order on **P**.

Example 4

Show that divisibility is **not** antisymmetric on **Z**.

Solution

2 | -2

because

2 × (-1) = -2

Also,

-2 | 2

because

(-2) × (-1) = 2

But

2 ≠ -2

Therefore,

Antisymmetry fails.

Hence,

Divisibility is **not** a partial order on **Z**.

Example 5

Let

S = {1,2,3}

Relation

R = {(1,1),(2,2),(3,3)}

Determine whether R is a partial order.

Solution

Reflexive ✓

Antisymmetric ✓

Transitive ✓

Therefore,

R is a partial order.

Example 6

Let

A = {a,b}

B = {a,b,c}

Find whether **A ⊆ B** defines an order.

Solution

A ⊆ A ✓

If A ⊆ B and B ⊆ A, then A = B ✓

Subset relation is transitive ✓

Hence,

Subset inclusion defines a partial order.

Example 7

Determine whether the relation

≤

on real numbers is a partial order.

Solution

Every real number satisfies

$a \le a$ ✓

If

$a \le b$

and

$b \le a$

then

$a = b$ ✓

If

$a \le b$

and

$b \le c$

then

$a \le c$ ✓

Hence,

$\le$ is a partial order on R.

Example 8

Consider

$2 \mid 6$

$6 \mid 18$

Show that divisibility is transitive.

Solution

2 divides 6

6 divides 18

Since

$18 = 2 \times 9$

Therefore,

2 divides 18.

Hence,

Transitivity is verified.

Example 9

Let

C = {∅ ,{1},{1,2}}

Verify reflexivity using subset inclusion.

Solution

$\emptyset \subseteq \emptyset$

$\{1\} \subseteq \{1\}$

$\{1,2\} \subseteq \{1,2\}$

Therefore,

Reflexivity holds.

Example 10

Determine whether the following relation on positive integers is a partial order.

Relation:

aRb if a divides b

Check:

3 | 12

12 | 24

Can we conclude

3 | 24?

Solution

Since

$12 = 3 \times 4$

and

$24 = 12 \times 2$

Therefore,

$24 = 3 \times 8$

Hence,

3 | 24

This verifies the **transitive property** of divisibility, confirming one of the essential conditions of a partial order.

LESSON 27: Comparable Elements, Linear Order and Chains (Summarized Notes)

In a partially ordered set **S**, let **a** and **b** be two distinct elements.

The elements **a** and **b** are called **comparable** if

a ≤ b or **b ≤ a**.

If neither **a ≤ b** nor **b ≤ a**, then **a** and **b** are called **non-comparable**.

A partially ordered set is called **linearly ordered** (or **totally ordered**) if **every pair of elements is comparable**.

In a linearly ordered set, every two elements can always be compared.

A partially ordered set may not be linearly ordered.

However, some of its subsets may still be linearly ordered.

A subset of a partially ordered set that is linearly ordered is called a **chain**.

Every pair of elements in a chain is comparable.

Example 1

The set of positive integers **P** ordered by **divisibility** is **not** linearly ordered because some elements are not comparable.

Example 2

The set of positive integers **P** with the usual order ≤ is **linearly ordered**.

Hence, every ordered subset of **P** is also linearly ordered.

Example 3

The power set **P(A)** of a set having two or more elements is **not** linearly ordered under **set inclusion (⊆)** because some subsets are not comparable.

Example 4

Let

A = {1,2,3,4,6,12,24}

ordered by divisibility.

The elements **3** and **4** are not comparable because

$3 \nmid 4$

and

$4 \nmid 3$

Therefore, **A** is **not** linearly ordered.

The following subsets are chains because every pair of elements is comparable.

{1,2,4,12,24}

{1,2,6,12,24}

{1,3,6,12,24}

Important Exam Points

Comparable elements satisfy

a ≤ b or **b ≤ a**.

Non-comparable elements satisfy neither relation.

A **linear (total) order** requires every pair of elements to be comparable.

A **chain** is a linearly ordered subset of a partially ordered set.

The positive integers with the usual order ≤ form a linear order.

The positive integers ordered by divisibility form a partial order but **not** a linear order.

The power set ordered by inclusion ⊆ is generally **not** linearly ordered.

Always verify **every pair of elements** when checking whether a subset is a chain.

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5 Very Important Statement-Based MCQs

1. Which of the following statements is TRUE about comparable elements in a partially ordered set?

A) Every pair of elements must be equal.

B) Two elements are comparable if **a ≤ b** or **b ≤ a**. ✅

C) Comparable elements must always be adjacent.

D) Comparable elements must belong to different sets.

---

2. Which statement correctly defines a linearly ordered set?

A) Only some pairs of elements are comparable.

B) Every pair of elements is comparable. ✅

C) No two elements are comparable.

D) Every element has exactly one predecessor.

---

3. Which of the following statements is CORRECT about a chain?

A) It is any subset of a partially ordered set.

B) It is a subset in which every pair of elements is comparable. ✅

C) It is a subset containing only one element.

D) It contains only maximal elements.

---

4. Which of the following sets is linearly ordered under the usual relation ≤?

A) Positive integers **P** ✅

B) Power set **P(A)** with two or more elements

C) Positive integers ordered by divisibility

D) Set of subsets ordered by inclusion

---

5. Why is the set {1,2,3,4,6,12,24} ordered by divisibility not linearly ordered?

A) Because 12 does not divide 24.

B) Because 1 does not divide every element.

C) Because 3 and 4 are not comparable. ✅

D) Because divisibility is not transitive.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {24,2,6}

ordered by divisibility.

Determine whether **A** is a chain.

Solution

2 | 6 ✓

6 | 24 ✓

2 | 24 ✓

Every pair is comparable.

Therefore,

A is a chain.

Example 2

Let

B = {3,15,5}

ordered by divisibility.

Determine whether **B** is a chain.

Solution

3 | 15 ✓

5 | 15 ✓

3 and 5 are not comparable.

Therefore,

Not every pair is comparable.

Hence,

B is not a chain.

Example 3

Let

C = {2,8,32,4}

ordered by divisibility.

Determine whether **C** is a chain.

Solution

2 | 4 ✓

4 | 8 ✓

8 | 32 ✓

Every pair is comparable.

Therefore,

C is a chain.

Example 4

Let

D = {7}

Determine whether **D** is a chain.

Solution

A single-element set has no incomparable pair.

Therefore,

D is a chain.

Example 5

Consider

A = {1,2,3,6}

ordered by divisibility.

Determine whether **A** is linearly ordered.

Solution

2 and 3 are not comparable.

$2 \nmid 3$

$3 \nmid 2$

Therefore,

A is not linearly ordered.

Example 6

Consider

A = {1,2,4,8,16}

ordered by divisibility.

Determine whether **A** is a chain.

Solution

1 | 2

2 | 4

4 | 8

8 | 16

Every earlier element divides every later element.

Hence,

A is a chain.

Example 7

Let

P(A) where

A = {1,2}

Consider the subsets

{1} and **{2}**.

Determine whether they are comparable under ⊆.

Solution

$\{1\} \not\subset \{2\}$

$\{2\} \not\subset \{1\}$

They are not comparable.

Hence,

P(A) is not linearly ordered.

Example 8

Consider the positive integers with the usual order ≤.

Check whether

5 and **12**

are comparable.

Solution

$5 \leq 12$

Therefore,

The two elements are comparable.

This supports that **(P, ≤)** is linearly ordered.

Example 9

Consider

A = {1,3,6,12,24}

ordered by divisibility.

Verify whether it is a chain.

Solution

1 | 3

3 | 6

6 | 12

12 | 24

Every pair is comparable through divisibility.

Therefore,

A is a chain.

Example 10

Consider

A = {1,2,3,4,6,12,24}

ordered by divisibility.

Determine whether **3** and **4** are comparable.

Solution

Check divisibility:

$3 \nmid 4$

$4 \nmid 3$

Neither element divides the other.

Therefore,

3 and 4 are non-comparable.

LESSON 28: Immediate Predecessor, Immediate Successor and Hasse Diagram (Summarized Notes)

An **immediate predecessor** and **immediate successor** describe elements that are directly connected in a partially ordered set.

Let **S** be a partially ordered set and let **a, b ∈ S**.

Element **a** is called the **immediate predecessor** of **b**.

Element **b** is called the **immediate successor** (or **cover**) of **a**.

This relation is written as

a << b

The relation **a << b** means

a < b

and

There is **no element c** in **S** such that

a < c < b

The symbol << represents the **cover relation**.

Example

For the positive integers ordered by <

1 << 2

2 << 3

3 << 4

Each number is the immediate successor of the previous one.

For a **finite partially ordered set**, the entire ordering can be determined by knowing all immediate predecessor-successor pairs.

If

x < y

then either

x << y

or there exists a sequence of immediate successors connecting **x** to **y**.

Divisibility Example

Let

A = {1,2,3,4,6,12,24}

ordered by divisibility.

The cover relations are

2 << 4

2 << 6

3 << 6

4 << 12

6 << 12

12 << 24

Some chains are

1 << 2 << 4 << 12 << 24

1 << 2 << 6 << 12 << 24

1 << 3 << 6 << 12 << 24

A **Hasse Diagram** is a graphical representation of a finite partially ordered set.

Each element is represented by a point.

A line is drawn between two elements whenever one is the immediate successor of the other.

Arrows are usually omitted by placing larger elements higher than smaller ones.

The Hasse Diagram displays only the cover relations.

It does **not** show transitive relations.

A Hasse Diagram can itself define a partially ordered set.

For the power set **P(A)** ordered by **subset inclusion (⊆)**

The empty set is placed at the bottom.

Singleton subsets appear on the first level.

Two-element subsets appear on the second level.

The complete set appears at the top.

Adjacent levels are connected using the cover relation.

To verify that a relation is a partial order, check the three properties.

Reflexive

Antisymmetric

Transitive

If all three properties hold, a Hasse Diagram can then be constructed.

Important Exam Points

Immediate predecessor is written as

a << b

The cover relation means

a < b

with **no element between them**.

Immediate successor and cover have the same meaning.

A Hasse Diagram represents only **immediate successor (cover) relations**.

Transitive edges are **never drawn** in a Hasse Diagram.

Higher elements are placed above lower elements.

Arrows are usually omitted.

The Hasse Diagram of a finite poset completely describes its ordering.

Always verify **Reflexive, Antisymmetric and Transitive** before drawing a Hasse Diagram.

For a power set ordered by ⊆

Empty set is at the bottom.

Universal (largest) set is at the top.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines the cover relation (a << b)?

A) (a = b)

B) (a < b) and there exists an element between them.

C) (a < b) and no element lies between them. ✅

D) (a > b)

---

2. Which statement about a Hasse Diagram is TRUE?

A) It contains all transitive edges.

B) It represents only immediate successor (cover) relations. ✅

C) It always uses arrows.

D) It represents only finite sets.

---

3. Which statement is CORRECT regarding a finite partially ordered set?

A) The ordering cannot be determined from cover relations.

B) Knowing all cover relations completely determines the ordering. ✅

C) Every finite partially ordered set is linearly ordered.

D) Hasse Diagrams cannot be drawn for finite sets.

---

4. In the Hasse Diagram of the power set ordered by inclusion, which subset is placed at the lowest level?

A) The complete set

B) Any singleton subset

C) The empty set (∅) ✅

D) Any two-element subset

---

5. Before drawing a Hasse Diagram for a relation, which properties must first be verified?

A) Symmetric, Reflexive and Complete

B) Reflexive, Antisymmetric and Transitive ✅

C) Symmetric and Transitive only

D) Reflexive and Symmetric only

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {1,2,4,8,16}

ordered by divisibility.

Find all cover relations.

Solution

1 << 2

2 << 4

4 << 8

8 << 16

No intermediate element exists between each pair.

These are all the cover relations.

Example 2

Consider

A = {1,2,3,6,18}

ordered by divisibility.

Determine whether

2 << 18

Solution

2 divides 18.

However,

2 << 6 << 18

Since **6** lies between **2** and **18**,

2 << 18 is **false**.

Example 3

Let

A = {1,2,3,6,12}

ordered by divisibility.

List all cover relations.

Solution

1 << 2

1 << 3

2 << 6

3 << 6

6 << 12

These are the only immediate successor pairs.

Example 4

Draw the levels of the Hasse Diagram for

P({a,b})

Solution

Level 3

{a,b}

Level 2

{a}   {b}

Level 1

∅

Join only adjacent subsets differing by one element.

Example 5

Let

P({a,b,c})

How many vertices does its Hasse Diagram contain?

Solution

Number of subsets

$= 2^3$

= 8

Therefore,

The Hasse Diagram contains **8 vertices**.

Example 6

For

A = {1,2,3,4,6,12,24}

determine whether

2 << 12

Solution

2 divides 12.

But

2 << 4 << 12

and

2 << 6 << 12

There are intermediate elements.

Therefore,

2 << 12 is **not** a cover relation.

Example 7

Given

R = {(1,1),(2,2),(1,2),(2,3),(3,3),(1,3)}

Determine whether **R** is transitive.

Solution

(1,2) and (2,3)

imply

(1,3)

Since (1,3) belongs to R,

Transitivity is satisfied.

Example 8

Given

R = {(1,1),(2,2),(1,2),(2,1)}

Determine whether **R** is antisymmetric.

Solution

(1,2) and (2,1)

are both present.

But

$1 \neq 2$

Therefore,

Antisymmetry fails.

Hence,

R is **not** a partial order.

Example 9

For

A = {1,2,3,4,6,12,24}

find a chain from **1** to **24** using cover relations.

Solution

One valid chain is

1 << 2 << 4 << 12 << 24

Another valid chain is

1 << 2 << 6 << 12 << 24

Another valid chain is

1 << 3 << 6 << 12 << 24

All are correct.

Example 10 (Critical VU Paper Pattern)

Let

A = {1,2,3,6,12,24}

ordered by divisibility.

Construct the Hasse Diagram by first identifying all cover relations.

Solution

Step 1

Identify immediate successors.

1 << 2

1 << 3

2 << 6

3 << 6

6 << 12

12 << 24

Step 2

Arrange elements from bottom to top.

Bottom

1

Next level

2 3

Next level

6

Next level

12

Top

24

Step 3

Join only the cover relations.

Do **not** draw transitive edges such as

$1 \rightarrow 6$,

$2 \rightarrow 12$,

or

$1 \rightarrow 24$.

LESSON 29: Minimal and Maximal Elements (Summarized Notes)

A **minimal element** is an element of a partially ordered set that has no smaller element preceding it.

Let **S** be a partially ordered set.

An element **a ∈ S** is called a **minimal element** if

$\mathbf{x \leq a \Rightarrow x = a}$

This means there is no element different from **a** that is smaller than **a**.

A partially ordered set may have **one**, **many**, or **no** minimal elements.

Example

Let

A = {1,2,3,4,6,8,9,12,18,24}

ordered by divisibility.

The only minimal element is

1

because every other element is divisible by **1**.

A **maximal element** is an element that has no greater element succeeding it.

An element **b ∈ S** is called a **maximal element** if

$\mathbf{b \leq x \Rightarrow x = b}$

This means there is no element different from **b** that is greater than **b**.

A finite partially ordered set always has **at least one minimal element and one maximal element**.

An infinite partially ordered set may have **no minimal or maximal elements**.

Example

The set of integers

Z

with the usual order ≤

has **no minimal element** and **no maximal element**.

The set of positive integers

P = {1,2,3,…}

with the usual order ≤

has

Minimal element = **1**

No maximal element.

For the power set

P(A)

ordered by **subset inclusion (⊆)**

The empty set

∅

is always the **minimal element**.

The set

A

itself is always the **maximal element**.

A **first (least) element** is an element that is less than or equal to every element of the set.

A **last (greatest) element** is an element that is greater than or equal to every element of the set.

Every first element is minimal.

Every last element is maximal.

However, a minimal element is **not necessarily** the first element.

Similarly, a maximal element is **not necessarily** the last element.

Important Exam Points

Minimal element satisfies

$\mathbf{x \le a \Rightarrow x = a}$

Maximal element satisfies

$\mathbf{a \le x \Rightarrow x = a}$

A finite partially ordered set always has at least one minimal and one maximal element.

An infinite partially ordered set may have neither minimal nor maximal elements.

The set **Z** with the usual order has **no minimal and no maximal element**.

The set **P** with the usual order has **minimal element = 1** and **no maximal element**.

For the power set ordered by inclusion

Minimal element = ∅

Maximal element = **A**

Every first element is minimal.

Every last element is maximal.

Minimal does **not** always mean first.

Maximal does **not** always mean last.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines a minimal element in a partially ordered set?

A) It is greater than every other element.

B) No element strictly precedes it. ✅

C) It is comparable with every element.

D) It is the largest element of the set.

---

2. Which statement is TRUE for every finite partially ordered set?

A) It has exactly one minimal element.

B) It has no maximal element.

C) It has at least one minimal and one maximal element. ✅

D) It is always linearly ordered.

---

3. Which of the following statements is CORRECT about the set of integers (Z) under the usual order?

A) It has a minimal element only.

B) It has a maximal element only.

C) It has neither a minimal nor a maximal element. ✅

D) It has both a minimal and a maximal element.

---

4. In the power set (P(A)) ordered by inclusion, which element is always minimal?

A) A

B) Any singleton subset

C) The empty set (∅) ✅

D) Any proper subset

---

5. Which statement about the first and minimal elements is TRUE?

A) Every minimal element is always the first element.

B) Every first element is minimal. ✅

C) First and minimal elements are unrelated.

D) A partially ordered set cannot have a first element.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {1,2,3,4,6,8,9,12,18,24}

ordered by divisibility.

Find all minimal elements.

Solution

Since **1** divides every element and no element other than **1** divides **1**,

the only minimal element is

{1}

Example 2

Using the same set

A = {1,2,3,4,6,8,9,12,18,24}

ordered by divisibility,

find all maximal elements.

Solution

No element of **A** is divisible by a larger element within the set except itself.

The elements

8, 9, 18, 24

have no greater multiples inside **A**.

Therefore,

Maximal elements = {8,9,18,24}

Example 3

Determine the minimal and maximal elements of

P({a,b})

ordered by ⊆.

Solution

Power set

{∅, {a}, {b}, {a,b}}

Minimal element

∅

Maximal element

{a,b}

Example 4

Consider

P = {1,2,3,…}

with the usual order ≤.

Find the minimal and maximal elements.

Solution

Smallest element is

1

Hence,

Minimal element = **1**

There is no largest positive integer.

Therefore,

No maximal element.

Example 5

Consider

Z

with the usual order ≤.

Determine the minimal and maximal elements.

Solution

For every integer, a smaller integer exists.

For every integer, a larger integer exists.

Therefore,

No minimal element.

No maximal element.

Example 6

Let

A = {2,3,5,30}

ordered by divisibility.

Find all minimal elements.

Solution

2 is not divisible by any other element.

3 is not divisible by any other element.

5 is not divisible by any other element.

30 is divisible by 2, 3 and 5.

Therefore,

Minimal elements

{2,3,5}

Example 7

Let

A = {2,3,5,30}

ordered by divisibility.

Find all maximal elements.

Solution

30 has no multiple inside the set.

Therefore,

Maximal element

{30}

Example 8

Determine whether the set

A = {∅ ,{a},{b},{a,b}}

ordered by inclusion has a first and last element.

Solution

∅ is a subset of every subset.

Hence,

First (least) element = **∅**

Every subset is contained in

{a,b}

Hence,

Last (greatest) element = **{a,b}**

Example 9

Consider

A = {1,2,4,8,16}

ordered by divisibility.

Find the minimal and maximal elements.

Solution

1 divides every element.

Therefore,

Minimal element = **1**

16 has no larger multiple inside the set.

Therefore,

Maximal element = **16**

Example 10 (Critical VU Paper Pattern)

Consider the Hasse Diagram of a finite partially ordered set having the cover relations

1 << 2

1 << 3

2 << 6

3 << 6

6 << 12

Identify the minimal element, maximal element, first element and last element.

Solution

The element with no predecessor is

1

Hence,

Minimal element = **1**

First (least) element = **1**

The element with no successor is

12

Hence,

Maximal element = **12**

Last (greatest) element = **12**

This is a common **Virtual University Midterm and Final** question, where students are asked to identify minimal, maximal, first, and last elements directly from a Hasse Diagram.

LESSON 30: Supremum and Infimum (Summarized Notes)

An element can serve as a **lower bound** or an **upper bound** for a subset of a partially ordered set.

Let **S** be a partially ordered set and let **A ⊆ S**.

Lower Bound

An element **n ∈ S** is called a **lower bound** of **A** if it precedes every element of **A**.

Mathematically,

For every y ∈ A, n ≤ y

A subset may have one, many, or no lower bounds.

Infimum

If a lower bound is greater than or equal to every other lower bound, then it is called the **infimum** (greatest lower bound).

It is denoted by

inf(A)

The infimum is the **greatest** among all lower bounds.

Upper Bound

An element **m ∈ S** is called an **upper bound** of **A** if it succeeds every element of **A**.

Mathematically,

For every x ∈ A, x ≤ m

A subset may have one, many, or no upper bounds.

Supremum

If an upper bound is smaller than or equal to every other upper bound, then it is called the **supremum** (least upper bound).

It is denoted by

sup(A)

The supremum is the **smallest** among all upper bounds.

Bounded Above

A subset is **bounded above** if it has at least one upper bound.

Bounded Below

A subset is **bounded below** if it has at least one lower bound.

Bounded Set

A subset is called **bounded** if it has **both** an upper bound and a lower bound.

Example

Let

S = {1,2,3,…,8}

and

A = {4,5,7}

The lower bounds, upper bounds, infimum and supremum are determined from the Hasse diagram.

The set of rational numbers **Q** can also be used to study upper bounds, lower bounds, supremum and infimum.

Important Exam Points

Lower bound satisfies

$n \le y$ for every $y \in A$.

Upper bound satisfies

$x \le m$ for every $x \in A$.

Infimum = Greatest Lower Bound (GLB).

Notation:

inf(A)

Supremum = Least Upper Bound (LUB).

Notation:

sup(A)

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A set is **bounded above** if it has an upper bound.

A set is **bounded below** if it has a lower bound.

A set is **bounded** if it has both upper and lower bounds.

Do not confuse:

Infimum ≠ Minimum

Supremum ≠ Maximum

The infimum and supremum are determined with respect to the given partial order.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines a lower bound of a subset A?

A) It is greater than every element of A.

B) It precedes every element of A. ✅

C) It belongs to every subset of A.

D) It is always an element of A.

---

2. Which statement is TRUE about the infimum of a subset?

A) It is the smallest upper bound.

B) It is the greatest lower bound. ✅

C) It is the largest element of the set.

D) It is always equal to the minimum element.

---

3. Which of the following statements correctly defines the supremum of a subset?

A) It is the least upper bound. ✅

B) It is the greatest upper bound.

C) It is the least lower bound.

D) It is always the maximum element.

---

4. A subset is called bounded if it

A) Has only an upper bound.

B) Has only a lower bound.

C) Has both an upper bound and a lower bound. ✅

D) Contains only finite elements.

---

5. Which statement is CORRECT?

A) Every supremum is necessarily the maximum element.

B) Every infimum is necessarily the minimum element.

C) A set bounded above has at least one upper bound. ✅

D) Every subset has both an infimum and a supremum.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

S = {1,2,3,4,5,6,7,8}

with the usual order

and

A = {4,5,7}

Find all lower bounds.

Solution

A lower bound must satisfy

$n \leq 4$, 5 and 7.

Hence,

Lower bounds

{1,2,3,4}

Example 2

Using the same set,

find all upper bounds of

A = {4,5,7}

Solution

An upper bound must satisfy

$4 \leq m$

$5 \leq m$

$7 \leq m$

Hence,

Upper bounds

{7,8}

Example 3

Using

A = {4,5,7}

find

inf(A).

Solution

Lower bounds are

{1,2,3,4}

The greatest of these is

4

Therefore,

inf(A) = 4

Example 4

Using

A = {4,5,7}

find

sup(A).

Solution

Upper bounds are

{7,8}

The least upper bound is

7

Therefore,

sup(A) = 7

Example 5

Determine whether

A = {4,5,7}

is bounded.

Solution

Lower bound exists.

Upper bound exists.

Therefore,

A is bounded.

Example 6

Consider

A = {2,4,8}

ordered by the usual order.

Find

inf(A)

and

sup(A).

Solution

Greatest lower bound

= **2**

Least upper bound

= **8**

Hence,

inf(A)=2

sup(A)=8

Example 7

Let

A = {3,5,6}

in

S = {1,2,3,4,5,6,7}

Find all lower bounds and upper bounds.

Solution

Lower bounds

{1,2,3}

Upper bounds

{6,7}

Therefore,

inf(A)=3

sup(A)=6

Example 8

Consider the subset

A = {1,2,3,4}

of positive integers.

Determine whether it is bounded above and bounded below.

Solution

Lower bound exists

= **1**

Upper bound exists

= **4**

Therefore,

The subset is bounded.

Example 9

Let

A = {5,6,7,8}

Find

inf(A)

and

sup(A)

under the usual order.

Solution

Lower bounds

{1,2,3,4,5}

Greatest lower bound

= **5**

Upper bound

= **8**

Hence,

inf(A)=5

sup(A)=8

Example 10 (Critical VU Paper Pattern)

A Hasse Diagram represents

S = {1,2,3,4,5,6,7,8}

and the subset

A = {4,5,7}.

Without listing all elements of **S**, determine:

Lower bounds of **A**

Upper bounds of **A**

inf(A)

sup(A)

Whether **A** is bounded.

Solution

Identify all elements preceding every member of **A**.

These form the lower bounds.

The greatest among them is

inf(A).

Identify all elements succeeding every member of **A**.

These form the upper bounds.

The least among them is

sup(A).

Since both lower and upper bounds exist,

A is bounded.

LESSON 31: Similar Ordered Sets (Order Isomorphism) (Summarized Notes)

A **similar mapping** is a one-to-one function that preserves the order between two partially ordered sets.

Let **X** and **Y** be partially ordered sets.

A function

f : X → Y

is called a **similar mapping** if it is **one-to-one (injective)** and preserves the order relation.

For every **a, b ∈ X**

a ≤ b in X if and only if f(a) ≤ f(b) in Y

The order of elements before and after mapping must remain exactly the same.

This condition is equivalent to preserving the ordering in both directions.

If the order changes after mapping, the function is **not** a similar mapping.

Similar Ordered Sets

Two ordered sets **X** and **Y** are called **order-isomorphic**, **isomorphic**, or **similar** if there exists a **one-to-one correspondence (bijection)** between them that preserves the order relation.

This is written as

X ≅ Y

The mapping that establishes this similarity is called an **order-isomorphism**.

Two similar ordered sets have the **same order structure**, even if their elements are different.

Example 1

To determine whether two ordered sets are similar, verify that there exists a one-to-one correspondence preserving every order relation.

If the ordering is preserved completely, the sets are order-isomorphic.

Example 2

Let

$P = \{1,2,3,\ldots\}$

and

$E = \{2,4,6,\ldots\}$

Define

$f(x) = 2x$

Every positive integer maps to a unique even positive integer.

The order is preserved because

If

$a \leq b$

then

$2a \leq 2b$

Therefore,

$P \cong E$

Example 3

Consider

$P = \{1,2,3,\ldots\}$

and

$A = \{-1,-2,-3,\ldots\}$

with the usual ordering.

These two sets are **not order-isomorphic**.

The mapping reverses the order instead of preserving it.

Therefore,

$P \ncong A$

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important Exam Points

A similar mapping must be **injective**.

It must preserve the order relation.

The condition is

$\mathbf{a \leq b \Leftrightarrow f(a) \leq f(b)}$

Two ordered sets are **order-isomorphic** if there exists a **one-to-one correspondence** preserving order.

Notation for similar ordered sets:

$\mathbf{X \cong Y}$

The mapping establishing similarity is called an **order-isomorphism**.

Two sets may have different elements but still be order-isomorphic if their ordering is identical.

The function

$\mathbf{f(x)=2x}$

is an order-isomorphism from positive integers to even positive integers.

Positive integers are **not** order-isomorphic to negative integers under the usual order.

Always check both

Injectivity

Order preservation

before concluding two ordered sets are similar.

---

5 Very Important Statement-Based MCQs

1. Which of the following statements correctly defines a similar mapping?

A) It is any onto function.

B) It is a one-to-one function that preserves the order relation. ✅

C) It is a many-to-one function.

D) It is a constant function.

---

2. Two ordered sets are called order-isomorphic if

A) They have the same number of elements only.

B) There exists a one-to-one correspondence preserving the order relation. ✅

C) They contain identical elements.

D) Their Hasse diagrams have the same number of vertices only.

---

3. Which statement is TRUE regarding the function (f(x)=2x) from positive integers to even positive integers?

A) It is not injective.

B) It reverses the order.

C) It is an order-isomorphism. ✅

D) It is a constant mapping.

---

4. Which notation represents two similar ordered sets?

A) (X \subseteq Y)

B) (X \cap Y)

C) (X \cong Y) ✅

D) (X \times Y)

---

5. Why are the positive integers and the negative integers (under the usual order) not order-isomorphic?

A) Because their cardinalities are different.

B) Because no injective function exists.

C) Because the usual ordering is not preserved. ✅

D) Because the positive integers are finite.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

P = {1,2,3,4}

and

E = {2,4,6,8}

Define

f(x)=2x

Determine whether **f** is an order-isomorphism.

Solution

The function is one-to-one.

For every

$a \leq b$

we have

$2a \leq 2b$.

The order is preserved.

Therefore,

f is an order-isomorphism.

Example 2

Let

P = {1,2,3}

and

A = {-1,-2,-3}

Define

f(x)=-x

Determine whether **f** is an order-isomorphism.

Solution

$1 \leq 2$

but

$-1 \geq -2$

The order is reversed.

Therefore,

f is not an order-isomorphism.

Example 3

Let

X = {1,2,3}

and

Y = {4,5,6}

Define

f(1)=4

f(2)=5

f(3)=6

Determine whether **X ≅ Y**.

Solution

The function is bijective.

The order is preserved.

Therefore,

X ≅ Y

Example 4

Let

X = {a,b,c}

with

a < b < c

and

Y = {2,4,6}

with

2 < 4 < 6

Construct an order-isomorphism.

Solution

a → 2

b → 4

c → 6

The order is preserved.

Hence,

The mapping is an order-isomorphism.

Example 5

Let

X = {1,2,3}

and

Y = {2,4,6}

Define

f(1)=6

f(2)=4

f(3)=2

Determine whether **f** is similar.

Solution

$1 < 2$

but

$6 > 4$

Order is not preserved.

Therefore,

f is not a similar mapping.

Example 6

Show that

f(x)=2x

is injective.

Solution

Suppose

2a = 2b

Dividing both sides by 2,

a = b

Hence,

f is one-to-one.

Example 7

Let

X = {1,2,3,4}

and

Y = {10,20,30,40}

Define

f(x)=10x

Determine whether the ordered sets are similar.

Solution

The mapping is one-to-one.

If

$a \leq b$

then

$10a \leq 10b$

The order is preserved.

Therefore,

X ≅ Y

Example 8

Determine whether the positive integers and even positive integers have the same order structure.

Solution

Using

f(x)=2x

every positive integer corresponds uniquely to an even positive integer.

The ordering remains unchanged.

Hence,

The two ordered sets are **order-isomorphic**.

Example 9

Let

X = {1,2,3}

and

Y = {1,3,2}

with the usual order of their listed elements.

Determine whether these ordered sets are similar.

Solution

The correspondence changes the order.

Order preservation fails.

Therefore,

The sets are **not order-isomorphic**.

Example 10 (Critical VU Paper Pattern)

Suppose

X = {a,b,c,d}

with

$a < b < c < d$

and

Y = {2,4,6,8}

ordered by the usual relation.

Construct an order-isomorphism and verify that it preserves the order.

Solution

Define

$a \rightarrow 2$

$b \rightarrow 4$

$c \rightarrow 6$

$d \rightarrow 8$

The mapping is one-to-one and onto.

For every pair of elements,

if

$a \leq b$

then

$f(a) \leq f(b)$

Similarly,

all remaining order relations are preserved.

Hence,

$X \cong Y$

LESSON 32: Well-Ordered Sets (Summarized Notes)

A **well-ordered set** is a special type of ordered set in which every subset has a first (least) element.

Let **A** be an ordered set.

The set **A** is called **well-ordered** if **every non-empty subset of A contains a first element**.

The first element is the least element of that subset.

Every well-ordered set is **linearly ordered**.

However, not every linearly ordered set is well-ordered.

If **A** is a well-ordered set, then the following properties hold.

Every subset of **A** is also well-ordered.

If another ordered set **B** is similar (order-isomorphic) to **A**, then **B** is also well-ordered.

An **initial segment** of an element **$a \in A$** consists of all elements that strictly precede **a**.

It is denoted by

s(a)

Mathematically,

$s(a) = \{x \in A : x < a\}$

The collection of all initial segments of **A** is denoted by

S(A)

The set **S(A)** is ordered by **set inclusion (⊆)**.

There exists a similarity mapping

$f : A \to S(A)$

defined by

$f(x) = s(x)$

This mapping preserves the order.

Therefore,

$A \cong S(A)$

Example

Consider

$A = \{1,2,3\}$

with the usual ordering.

The initial segments are

$s(1) = \emptyset$

$s(2) = \{1\}$

$s(3) = \{1,2\}$

Thus,

$S(A) = \{\emptyset,\{1\},\{1,2\}\}$

The mapping

$f(x)=s(x)$

is an order-isomorphism.

Important Exam Points

A well-ordered set is an ordered set in which **every non-empty subset has a first element**.

Every well-ordered set is **linearly ordered**.

Every subset of a well-ordered set is also well-ordered.

If **A** ≅ **B** and **A** is well-ordered, then **B** is also well-ordered.

The initial segment of an element **a** is

s(a) = {x ∈ A : x < a}

The collection of all initial segments is denoted by

S(A)

The mapping

f(x)=s(x)

is a similarity mapping.

Therefore,

A ≅ S(A)

Be able to construct initial segments for a given ordered set.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines a well-ordered set?

A) Every subset has a maximal element.

B) Every non-empty subset has a first element. ✅

C) Every subset is finite.

D) Every element has an immediate successor.

---

2. Which of the following statements is TRUE about every well-ordered set?

A) It is partially ordered only.

B) It is linearly ordered. ✅

C) It is finite.

D) It is unordered.

---

3. Which statement is CORRECT if A is a well-ordered set?

A) Some subsets of A may not be well-ordered.

B) Every subset of A is well-ordered. ✅

C) Only finite subsets are well-ordered.

D) Only singleton subsets are well-ordered.

---

4. The initial segment (s(a)) of an element (a) consists of

A) All elements greater than a.

B) All elements equal to a.

C) All elements strictly preceding a. ✅

D) All incomparable elements.

---

5. Which mapping establishes the similarity between (A) and (S(A))?

A) (f(x)=x)

B) (f(x)=A)

C) (f(x)=s(x)) ✅

D) (f(x)=x+1)

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {1,2,3,4}

with the usual order.

Determine whether **A** is well-ordered.

Solution

Every non-empty subset has a least element.

Therefore,

A is well-ordered.

Example 2

Find the initial segments of

A = {1,2,3,4}

Solution

s(1)=∅

s(2)={1}

s(3)={1,2}

s(4)={1,2,3}

Example 3

Construct

S(A)

for

A = {1,2,3}

Solution

s(1)=∅

s(2)={1}

s(3)={1,2}

Therefore,

S(A)={∅,{1},{1,2}}

Example 4

Define the similarity mapping

between

A={1,2,3}

and

S(A).

Solution

f(1)=∅

f(2)={1}

f(3)={1,2}

The mapping preserves the order.

Hence,

A ≅ S(A)

Example 5

Consider the subset

B={2,3,4}

of

A={1,2,3,4}

Determine whether **B** is well-ordered.

Solution

Least element of B is

2

Every non-empty subset of B also has a least element.

Therefore,

B is well-ordered.

Example 6

Let

A={1,2,3,4,5}

Find

s(5).

Solution

All elements preceding 5 are

{1,2,3,4}

Therefore,

s(5)={1,2,3,4}

Example 7

Let

A={1,2,3,4,5}

Find

s(1).

Solution

No element precedes 1.

Therefore,

s(1)=∅

Example 8

Suppose

A ≅ B

and

A is well-ordered.

Determine whether B is well-ordered.

Solution

A theorem states that order-isomorphic sets preserve well-ordering.

Therefore,

B is well-ordered.

Example 9

Let

A={1,2,3,4}

Arrange the elements of

S(A)

according to inclusion.

Solution

∅

⊆ {1}

$\subseteq \{1,2\}$

$\subseteq \{1,2,3\}$

This forms a well-ordered chain under inclusion.

Example 10 (Critical VU Paper Pattern)

Let

$\mathbf{A=\{1,2,3\}}$

with the usual ordering.

Construct all initial segments, form

$\mathbf{S(A)}$,

define the mapping

$\mathbf{f(x)=s(x)}$,

and verify that it is an order-isomorphism.

Solution

Initial segments:

$s(1)=\emptyset$

$s(2)=\{1\}$

$s(3)=\{1,2\}$

Hence,

$\mathbf{S(A)=\{\emptyset,\{1\},\{1,2\}\}}$

Define

$f(1)=\emptyset$

$f(2)=\{1\}$

$f(3)=\{1,2\}$

The mapping is one-to-one.

For every pair

$1<2<3$

we obtain

$\emptyset \subset \{1\} \subset \{1,2\}$

The order is preserved.

Therefore,

f is an **order-isomorphism**, and

A ≅ S(A).

LESSON 33: Transfinite Induction and Limit Elements (Summarized Notes)

Transfinite induction is an extension of ordinary mathematical induction for **well-ordered sets**.

It is used to prove statements about all elements of a well-ordered set.

Principle of Transfinite Induction

Let **S** be a subset of a well-ordered set **A**.

To prove **S = A**, the following conditions must hold.

The first element of **A** belongs to **S**.

Whenever all elements preceding an element belong to **S**, then that element also belongs to **S**.

If these conditions are satisfied, every element of the well-ordered set belongs to **S**.

Therefore,

S = A

Transfinite induction generalizes the ordinary principle of mathematical induction.

Limit Elements

An element **a** in a well-ordered set is called a **limit element** if

It is **not the first element**.

It **does not have an immediate predecessor**.

A limit element cannot be obtained by moving one step forward from another element.

Finite well-ordered sets generally do **not** contain limit elements because every element except the first has an immediate predecessor.

Limit elements mainly occur in infinite well-ordered sets.

Comparison of Well-Ordered Sets

Let **A** and **B** be well-ordered sets.

If **A** is similar (order-isomorphic) to an **initial segment** of **B**, then

A is said to be shorter than B.

Similarly,

B is said to be longer than A.

The comparison depends on the **order structure**, not on the actual elements.

Order-isomorphism is used to compare the lengths of well-ordered sets.

Important Exam Points

Transfinite induction is an extension of mathematical induction.

It is applicable only to **well-ordered sets**.

To apply transfinite induction,

The first element must belong to the subset.

Every element belongs whenever all preceding elements belong.

A **limit element** has **no immediate predecessor**.

A limit element is **not the first element**.

Finite well-ordered sets usually have **no limit elements**.

If **A** is similar to an initial segment of **B**, then

A is shorter than B.

The comparison of well-ordered sets is based on **order-isomorphism**.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly describes the Principle of Transfinite Induction?

A) It applies only to finite sets.

B) It extends mathematical induction to well-ordered sets. ✅

C) It applies only to linearly ordered sets.

D) It is used only for partially ordered sets.

---

2. Which statement is TRUE about a limit element?

A) It is always the first element.

B) It has an immediate predecessor.

C) It has no immediate predecessor and is not the first element. ✅

D) It exists in every finite ordered set.

---

3. Which of the following sets generally contains limit elements?

A) Finite well-ordered sets.

B) Infinite well-ordered sets. ✅

C) Empty sets only.

D) Singleton sets only.

---

4. Which statement correctly compares two well-ordered sets A and B?

A) A is shorter than B if A has fewer elements.

B) A is shorter than B if A is similar to an initial segment of B. ✅

C) A is shorter than B if every element of A belongs to B.

D) A is shorter than B only when A is finite.

---

5. Which property is essential for applying transfinite induction?

A) The set must be finite.

B) The set must be well-ordered. ✅

C) The set must be countable.

D) The set must contain limit elements.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether

A = {1,2,3,4,5}

with the usual order is well-ordered.

Can transfinite induction be applied?

Solution

Every non-empty subset has a first element.

Therefore,

A is well-ordered.

Hence,

Transfinite induction can be applied.

Example 2

Consider

A = {1,2,3,4}

Identify the first element.

Solution

The least element is

1

This is the base element required in transfinite induction.

Example 3

Consider

A = {1,2,3,4,5}

Determine whether **4** is a limit element.

Solution

Immediate predecessor of 4 is

3

Therefore,

4 has an immediate predecessor.

Hence,

4 is not a limit element.

Example 4

Determine whether finite well-ordered sets contain limit elements.

Solution

Every element except the first has an immediate predecessor.

Therefore,

Finite well-ordered sets contain **no limit elements**.

Example 5

Suppose

A ≅ {1,2,3}

which is an initial segment of

B ≅ {1,2,3,4,5}

Compare A and B.

Solution

Since A is similar to an initial segment of B,

A is shorter than B

and

B is longer than A.

Example 6

Let

A = {1,2,3}

and

B = {1,2,3,4}

Determine whether A is shorter than B.

Solution

A is identical to the initial segment

{1,2,3}

of B.

Therefore,

A is shorter than B.

Example 7

Suppose every element preceding **x** belongs to subset **S**, and **x** also belongs to **S**.

Which principle is being applied?

Solution

This is the inductive step of

Transfinite Induction.

Example 8

Let

A = {1,2,3,4,5,6}

Suppose

S = {1,2,3,4,5,6}

after applying the base case and induction step.

Determine the relationship between **S** and **A**.

Solution

All elements of A belong to S.

Therefore,

S = A

by the Principle of Transfinite Induction.

Example 9

Suppose

A is not well-ordered.

Can the Principle of Transfinite Induction be applied?

Solution

No.

Transfinite induction requires the underlying set to be **well-ordered**.

Therefore,

The principle cannot be applied.

Example 10 (Critical VU Paper Pattern)

Let

A = {1,2,3,4,5}

be a well-ordered set.

A subset **S** satisfies the following conditions:

The first element **1** belongs to **S**.

Whenever all elements preceding an element belong to **S**, that element also belongs to **S**.

Use the Principle of Transfinite Induction to determine **S**.

Solution

Base Step:

$1 \in S$

Induction Step:

Since every element is included whenever all preceding elements are included,

$2 \in S$

$3 \in S$

$4 \in S$

$5 \in S$

Hence,

Every element of **A** belongs to **S**.

Therefore,

S = A

LESSON 34: Choice Function, Axiom of Choice, Well-Ordering Theorem (Summarized Notes)

A **choice function** is a function that selects exactly one element from every non-empty set in a collection of non-empty sets.

Let **C** be a non-empty set of non-empty sets.

A function

$\mathbf{f : C \to \cup C}$

is called a **choice function** if

$\mathbf{f(A) \in A}$

for every

$\mathbf{A \in C}$

This means the function chooses one element from each set.

Examples of Choice Functions

If **C** is the set of all countries, viewed as collections of cities, then the union of C is the set of all cities.

A function that assigns each country its capital city is a choice function.

For example,

If

A = Pakistan

then

f(A) = Islamabad ∈ A

If **C** is the collection of all pairs of shoes, then selecting the **left shoe** from every pair is also a choice function.

If

C = P(N) − {∅}

then

f(A) = min(A)

is a choice function because every non-empty subset of natural numbers has a smallest element.

Axiom of Choice

Sometimes a choice function can be defined by a simple rule such as choosing the smallest element, the left shoe, or the capital city.

However, for many collections there may be **no obvious rule** for selecting one element from each set.

The **Axiom of Choice** guarantees that a choice function always exists.

Statement of the Axiom of Choice

Let **C** be a non-empty set of non-empty sets.

Then there exists a choice function for **C**.

The axiom guarantees existence of the function even if we cannot explicitly define it.

Well-Ordered Set

A set is called **well-ordered** if every non-empty subset has a **least element**.

The set of positive integers with the usual order is well-ordered.

The set of even positive integers is also well-ordered.

The set

{1,5,12,17}

with the usual order is well-ordered because every non-empty subset has a smallest element.

The set of integers **Z** with the usual order is **not well-ordered** because subsets such as negative integers have no least element.

The set of rational numbers is also **not well-ordered** under the usual order.

Whether a set is well-ordered depends on the ordering defined on that set.

Well-Ordering Theorem

The **Well-Ordering Theorem** states that

Every set can be well-ordered.

Even though the integers are not well-ordered under the usual order, another ordering can be defined that makes them well-ordered.

One such ordering is based on absolute values.

For integers

$\mathbf{a \prec b}$

whenever

$|\mathbf{a}| < |\mathbf{b}|$

or

$|\mathbf{a}| = |\mathbf{b}|$

and

a is negative while **b** is positive.

Thus every set can be arranged into a well-ordered form.

Zorn's Lemma

Let **X** be a non-empty partially ordered set.

If every chain (linearly ordered subset) of **X** has an upper bound in **X**, then **X** contains at least one maximal element.

For example,

Consider

X = {1,2,3,4,6,12,24}

ordered by divisibility.

Every chain has the upper bound

24

Therefore,

24 is a maximal element of **X**.

Equivalence of Three Fundamental Results

The following three statements are mathematically equivalent.

Axiom of Choice.

Well-Ordering Theorem.

Zorn's Lemma.

If one of them is true, then the other two are also true.

Ordinal Numbers

The ordinal numbers of the well-ordered sets

∅, {1}, {1,2}, {1,2,3},...

are denoted by

0,1,2,3,...

These are called **finite ordinal numbers**.

Any ordinal larger than these is called a **transfinite ordinal number**.

The ordinal number of the set of counting numbers is denoted by

ω (omega)

Comparison of Ordinal Numbers

Let

λ = ord(A)

and

μ = ord(B)

where **A** and **B** are well-ordered sets.

Then

$\lambda < \mu$

if **A** is similar to an initial segment of **B**.

This comparison depends on the order structure of the sets.

Important Exam Points

A choice function selects one element from every non-empty set.

For every set **A**, the chosen element satisfies **$f(A) \in A$**.

The Axiom of Choice guarantees the existence of a choice function.

A choice function may exist even when no explicit rule is known.

A well-ordered set is one in which every non-empty subset has a least element.

Positive integers are well-ordered under the usual order.

Integers and rational numbers are not well-ordered under the usual order.

The Well-Ordering Theorem states that every set can be well-ordered.

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Zorn's Lemma states that every partially ordered set whose chains have upper bounds contains a maximal element.

The Axiom of Choice, Well-Ordering Theorem, and Zorn's Lemma are equivalent.

Ordinal numbers describe the order type of well-ordered sets.

The ordinal number of the counting numbers is **ω (omega)**.

Finite ordinals are **0,1,2,3,...**

All larger ordinals are transfinite ordinals.

Ordinal comparison is based on similarity with an initial segment.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines a choice function?

A) It selects every element from each set.

B) It selects one element from every non-empty set. ✅

C) It selects only the largest element.

D) It selects only finite sets.

---

2. What does the Axiom of Choice guarantee?

A) Every set is finite.

B) Every set is countable.

C) A choice function exists for every non-empty collection of non-empty sets. ✅

D) Every set has a maximum element.

---

3. Which statement is TRUE about a well-ordered set?

A) Every subset has a greatest element.

B) Every non-empty subset has a least element. ✅

C) Every element has a predecessor.

D) Every subset is finite.

---

4. Which theorem states that every set can be well-ordered?

A) Cantor's Theorem.

B) Well-Ordering Theorem. ✅

C) De Morgan's Law.

D) Schröder-Bernstein Theorem.

---

5. Which three mathematical statements are equivalent?

A) Inclusion Principle, Pigeonhole Principle, De Morgan's Law.

B) Axiom of Choice, Well-Ordering Theorem, and Zorn's Lemma. ✅

C) Mathematical Induction, Contradiction, and Contrapositive.

D) Reflexive, Symmetric, and Transitive Properties.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether

C = {{1,2},{3,4},{5,6}}

admits a choice function.

Solution

Choose

1 from {1,2}

Choose

3 from {3,4}

Choose

5 from {5,6}

Each chosen element belongs to its respective set.

Therefore,

A choice function exists.

Example 2

Let

C = P(N) − {∅}

Define

f(A)=min(A)

Is **f** a choice function?

Solution

Every non-empty subset of natural numbers has a minimum element.

Also,

min(A) ∈ A

Therefore,

f(A)=min(A) is a valid choice function.

Example 3

Determine whether

{2,4,6,8,...}

with the usual order is well-ordered.

Solution

Every non-empty subset has a least even number.

Therefore,

The set is well-ordered.

Example 4

Determine whether the integers **Z** with the usual order are well-ordered.

Solution

The subset

{...,−3,−2,−1}

has no least element.

Therefore,

The integers are **not** well-ordered.

Example 5

State whether the positive rational numbers are well-ordered.

Solution

There is no least element in subsets such as

{1/2,1/3,1/4,...}

Therefore,

Positive rational numbers are not well-ordered.

Example 6

Consider

X={1,2,3,4,6,12,24}

ordered by divisibility.

Find the maximal element.

Solution

Every chain has upper bound

24

No element is greater than 24 under divisibility.

Therefore,

The maximal element is

24.

Example 7

Which theorem guarantees that every set can be arranged into a well-ordered form?

Solution

The theorem is the

Well-Ordering Theorem.

Therefore,

Every set can be well-ordered.

Example 8

Name the three equivalent mathematical principles.

Solution

Axiom of Choice.

Well-Ordering Theorem.

Zorn's Lemma.

Therefore,

Each statement implies the other two.

Example 9

Find the ordinal number of the set of counting numbers.

Solution

The ordinal number of the counting numbers is

ω (omega).

Therefore,

The first infinite ordinal is

ω.

Example 10 (Critical VU Paper Pattern)

Let

A and **B**

be well-ordered sets with

ord(A)=λ

and

ord(B)=μ

Suppose

A is similar to an initial segment of **B**.

Compare **λ** and **μ**.

Solution

If **A** is similar to an initial segment of **B**,

then the order type of **A** is smaller than the order type of **B**.

Therefore,

$\lambda < \mu$

LESSON 35: Propositions, Compound Propositions and Logical Operations (Summarized Notes)

A **proposition** is a declarative statement that has exactly one truth value.

A proposition is either **True (T)** or **False (F)**, but it cannot be both at the same time.

Only declarative sentences are considered propositions.

Questions, commands, and open sentences containing variables are **not** propositions unless the variables are assigned values.

Examples of Propositions

The following are propositions because they are either true or false.

Paris is in France.

1 + 1 = 2.

2 + 2 = 3.

Islamabad is in Sri Lanka.

9 < 6.

The following are **not** propositions.

Where are you going?

Do your homework.

$x^2 = 4$

This is not a proposition because the truth value depends on the value of **x**.

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Propositional Variables

Letters are used to represent propositions.

The commonly used propositional variables are

p, q, r, s, ...

Each propositional variable has one of two truth values.

T represents **True**.

F represents **False**.

Compound Propositions

A **compound proposition** is formed by combining two or more simple propositions using logical operators.

Examples of compound propositions are

Roses are red and violets are blue.

John is intelligent or studies every night.

Logical operators are used to connect propositions and produce new propositions.

Conjunction (AND)

The conjunction of two propositions is formed using the word **and**.

It is denoted by

p ∧ q

and is read as

p and q

The conjunction is **True** only when both propositions are true.

If either proposition is false, then the conjunction is false.

Truth Table of Conjunction

p q p ∧ q

p q p ∧ q
T T T
T F F
F T F
F F F

Disjunction (OR)

The disjunction of two propositions is formed using the word **or**.

It is denoted by

p ∨ q

and is read as

p or q

The disjunction is **False** only when both propositions are false.

If at least one proposition is true, then the disjunction is true.

Truth Table of Disjunction

p q p ∨ q
T T T
T F T
F T T
F F F

Negation (NOT)

The negation of a proposition changes its truth value.

It is denoted by

~p

and is read as

not p

The truth value of **~p** is always the opposite of the truth value of **p**.

Truth Table of Negation

p	**~p**
T	F
F	T

Important Exam Points

A proposition is a declarative sentence with exactly one truth value.

Every proposition is either **True** or **False**.

A proposition cannot be both true and false at the same time.

Questions and commands are not propositions.

Statements containing variables are not propositions unless the variables are assigned values.

A compound proposition is formed by combining simple propositions.

The conjunction operator is represented by ∧.

The conjunction is true only when both propositions are true.

The disjunction operator is represented by ∨.

The disjunction is false only when both propositions are false.

The negation operator is represented by ~.

Negation changes a true proposition into false and a false proposition into true.

The propositional variables commonly used are **p, q, r, s**.

---

5 Very Important Statement-Based MCQs

1. Which statement correctly defines a proposition?

A) A sentence that is always true.

B) A declarative sentence that is either true or false. ✅

C) A question with two answers.

D) A command given to someone.

---

2. Which of the following is NOT a proposition?

A) Paris is in France.

B) $9 < 6$.

C) Where are you going? ✅

D) $2 + 2 = 3$.

---

3. When is the conjunction $p \wedge q$ true?

A) When at least one proposition is true.

B) Only when both propositions are true. ✅

C) Only when both propositions are false.

D) Whenever p is true.

---

4. When is the disjunction $p \vee q$ false?

A) When p is true.

B) When q is true.

C) When both propositions are false. ✅

D) When both propositions are true.

---

5. What is the truth value of ~p if p is true?

A) True.

B) False. ✅

C) Both true and false.

D) Cannot be determined.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether the statement

"5 is greater than 3."

is a proposition.

Solution

The statement is declarative.

Its truth value is **True**.

Therefore,

It is a proposition.

Example 2

Determine whether

"Do your homework."

is a proposition.

Solution

This is a command.

Commands do not have truth values.

Therefore,

It is **not** a proposition.

Example 3

Determine whether

x + 2 = 5

is a proposition.

Solution

The truth value depends on the value of **x**.

Since **x** is not specified,

the statement has no definite truth value.

Therefore,

It is **not** a proposition.

Example 4

Let

p = True

and

q = False

Find

p ∧ q

Solution

A conjunction is true only when both propositions are true.

Here,

p = T

q = F

Therefore,

p ∧ q = False.

Example 5

Let

p = True

and

q = False

Find

p ∨ q

Solution

A disjunction is true if at least one proposition is true.

Since **p** is true,

Therefore,

p ∨ q = True.

Example 6

If

p = False

find

~p

Solution

Negation changes the truth value.

Since

p = False,

Therefore,

~p = True.

Example 7

Construct the conjunction of the following propositions.

p: Ali is a student.

q: Ali plays cricket.

Solution

The conjunction is

Ali is a student and Ali plays cricket.

Symbolically,

$p \land q$

Example 8

Construct the disjunction of the following propositions.

p: It is raining.

q: It is windy.

Solution

The disjunction is

It is raining or it is windy.

Symbolically,

$p \lor q$

Example 9

Let

p = False

and

q = False

Find

p ∨ q

and

p ∧ q

Solution

Since both propositions are false,

p ∨ q = False

Also,

p ∧ q = False

Example 10 (Critical VU Paper Pattern)

Let

p = True

and

q = True

Determine the truth values of

p ∧ q

p ∨ q

~p

Solution

Since both propositions are true,

p ∧ q = True

At least one proposition is true.

Therefore,

p ∨ q = True

The negation of a true proposition is false.

Therefore,

~p = False

LESSON 36: More Logical Operations and Truth Table of Compound Propositions (Summarized Notes)

A **conditional statement** is a compound proposition formed by connecting two propositions using the words **if...then**.

If **p** and **q** are propositions, then

p → q

is read as

If p, then q.

In a conditional statement,

p is called the **hypothesis**.

q is called the **conclusion**.

The conditional statement is **False** only when the hypothesis is true and the conclusion is false.

In every other case, the conditional statement is **True**.

Truth Table of Conditional Statement

p q p → q
T T T
T F F

p q p → q
F T T
F F T

The only false case occurs when **p is true** and **q is false**.

Understanding Conditional Statements

Consider the statement

If you get 100% on the final, then you will get an A.

If the student gets **100%** but does **not** get an A, then the statement is false.

In all other situations, the statement is considered true.

Different Ways to Express a Conditional Statement

The statement

p → q

can be written in different forms.

If p, then q.

p implies q.

q whenever p.

p is sufficient for q.

q if p.

q when p.

All of these statements have the same logical meaning.

Logical Equivalence of Conditional Statement

The conditional statement

p → q

is logically equivalent to

$\sim p \vee q$

This means both statements always have the same truth values.

Therefore,

$p \rightarrow q \equiv \sim p \vee q$

This is one of the most important logical equivalences in discrete mathematics.

Example of Conditional Statement

Let

p: Maria learns discrete mathematics.

q: Maria will find a good job.

Then

$p \rightarrow q$

means

If Maria learns discrete mathematics, then she will find a good job.

It can also be written as

Maria will find a good job when she learns discrete mathematics.

Both statements have the same meaning.

Compound Propositions

Compound propositions are formed by combining simple propositions using logical connectives.

The main logical connectives are

Negation ($\sim$)

Conjunction ($\wedge$)

Disjunction ($\vee$)

Conditional ($\rightarrow$)

These connectives allow us to construct complex logical statements.

Truth Tables of Compound Propositions

A **truth table** lists all possible truth values of propositions and determines the truth value of a compound proposition.

Truth tables are used to

Determine the truth value of compound propositions.

Verify logical equivalence.

Analyze logical arguments.

Simplify logical expressions.

Important Exam Points

A conditional statement is represented by **$p \to q$**.

The statement **p** is called the hypothesis.

The statement **q** is called the conclusion.

A conditional statement is false only when **p is true** and **q is false**.

In all other cases, a conditional statement is true.

The statement **$p \to q$** is logically equivalent to **$\sim p \lor q$**.

Logical equivalence means two propositions always have identical truth values.

Truth tables are used to determine the truth values of compound propositions.

The main logical connectives are **$\sim$, $\land$, $\lor$, $\to$**.

A truth table considers every possible combination of truth values.

---

5 Very Important Statement-Based MCQs

1. When is the conditional statement $p \to q$ false?

A) When both p and q are true.

B) When p is false and q is true.

C) When p is true and q is false. ✅

D) When both p and q are false.

---

2. In the conditional statement p → q, p is called

A) Conclusion.

B) Negation.

C) Hypothesis. ✅

D) Disjunction.

---

3. Which statement is logically equivalent to p → q?

A) p ∧ q

B) p ∨ q

C) ~p ∨ q ✅

D) ~p ∧ q

---

4. Which logical operator represents a conditional statement?

A) ∧

B) ∨

C) →

D) → ✅

---

5. What is the main purpose of a truth table?

A) To count propositions.

B) To determine the truth values of compound propositions. ✅

C) To simplify arithmetic expressions.

D) To solve equations.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

p = True

and

q = True

Find

p → q

Solution

A conditional statement is false only when

p = True

and

q = False.

Here,

p = True

q = True

Therefore,

p → q = True

Example 2

Let

p = True

and

q = False

Find

$p \rightarrow q$

Solution

The hypothesis is true and the conclusion is false.

Therefore,

$p \rightarrow q$ = False

Example 3

Let

p = False

and

q = True

Find

$p \rightarrow q$

Solution

The hypothesis is false.

Therefore,

The conditional statement is true.

Hence,

$p \to q$ = True

Example 4

Let

p = False

and

q = False

Find

$p \to q$

Solution

Since the hypothesis is false,

the conditional statement is true.

Therefore,

$p \to q$ = True

Example 5

Let

p: It is raining.

q: The ground is wet.

Write the conditional statement.

Solution

The conditional statement is

If it is raining, then the ground is wet.

Symbolically,

$p \rightarrow q$

Example 6

Express the statement

$p \rightarrow q$

in another equivalent form.

Solution

Using logical equivalence,

$p \rightarrow q \equiv \sim p \vee q$

Therefore,

The equivalent statement is

Not p or q.

Example 7

Let

p: Ali studies.

q: Ali passes the exam.

Identify the hypothesis and conclusion.

Solution

Hypothesis:

Ali studies.

Conclusion:

Ali passes the exam.

Therefore,

The hypothesis is **p** and the conclusion is **q**.

Example 8

Construct the truth table for

~p ∨ q

when

p = True

and

q = False

Solution

Since

p = True,

~p = False

Also,

q = False

Therefore,

~p ∨ q = False

This agrees with the truth value of

p → q

Example 9

Determine whether

p → q

and

$\sim p \lor q$

are logically equivalent.

Solution

Their truth tables are identical for every possible combination of truth values.

Therefore,

$p \rightarrow q \equiv \sim p \lor q$

Example 10 (Critical VU Paper Pattern)

Let

p = True

and

q = False

Determine the truth values of

$p \rightarrow q$

and

$\sim p \lor q$

Verify whether they are logically equivalent.

Solution

Since

p = True

and

q = False,

$p \rightarrow q$ = False

Also,

$\sim p$ = False

Therefore,

$\sim p \lor q$ = False $\lor$ False = False

Both expressions have the same truth value.

Therefore,

$p \rightarrow q \equiv \sim p \lor q$

LESSON 37: Real Numbers and the Power of Continuum (Part 1) (Summarized Notes)

The **unit interval** is the set of all real numbers between **0 and 1**, including both endpoints.

It is denoted by

$I = [0,1]$

The unit interval contains infinitely many real numbers.

Unlike the set of natural numbers, the elements of the unit interval **cannot be counted one by one**.

Nondenumerable Set

A set is called **nondenumerable (uncountable)** if its elements cannot be placed into one-to-one correspondence with the natural numbers.

This means the elements of the set **cannot be listed in a sequence**

1st, 2nd, 3rd, ...

The unit interval

$I = [0,1]$

is a **nondenumerable set**.

This important result was proved by **Cantor's Diagonal Argument**.

Therefore,

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The number of real numbers between **0** and **1** is greater than the number of natural numbers.

Power of the Continuum

A set **A** is said to have the **power of the continuum** if it has the same cardinality as the unit interval

I = [0,1]

Symbolically,

|A| = |I|

A set having the power of the continuum has **uncountably many elements**.

Its size is larger than any countably infinite set.

Cardinality

Cardinality is the measure of the number of elements in a set.

Finite sets have finite cardinality.

Infinite sets may be

Countably infinite.

Uncountably infinite.

The unit interval has **uncountable cardinality**.

This cardinality is called the **power of the continuum**.

Real Numbers in the Unit Interval

The interval

[0,1]

contains

Whole numbers.

Decimal numbers.

Rational numbers.

Irrational numbers.

Since irrational numbers are infinitely many and cannot be counted, the interval becomes uncountable.

Importance of the Power of the Continuum

The power of the continuum is one of the most important concepts in set theory.

It shows that **not all infinite sets have the same size**.

The set of real numbers is strictly larger than the set of natural numbers.

This discovery was made by **Georg Cantor** and forms the foundation of modern set theory.

Important Exam Points

The unit interval is denoted by **I = [0,1]**.

The unit interval contains all real numbers from **0 to 1**, including both endpoints.

The unit interval is a **nondenumerable (uncountable)** set.

A nondenumerable set cannot be placed into one-to-one correspondence with the natural numbers.

The unit interval cannot be completely listed.

A set has the **power of the continuum** if it has the same cardinality as **[0,1]**.

The cardinality of the continuum is greater than the cardinality of the natural numbers.

The power of the continuum represents an uncountably infinite set.

Cantor proved that the unit interval is uncountable.

Not all infinite sets have the same cardinality.

---

5 Very Important Statement-Based MCQs

1. Which of the following is the unit interval?

A) (0,1)

B) [0,1] ✅

C) (1,∞)

D) [1,∞)

---

2. The unit interval [0,1] is

A) Finite.

B) Countably infinite.

C) Nondenumerable. ✅

D) Empty.

---

3. A set has the power of the continuum if it has the same cardinality as

A) The set of integers.

B) The set of natural numbers.

C) The unit interval **[0,1]**. ✅

D) The set of rational numbers.

---

4. Which mathematician proved that the unit interval is uncountable?

A) Euclid.

B) Euler.

C) Cantor. ✅

D) Gauss.

---

5. Which statement is TRUE?

A) All infinite sets have the same cardinality.

B) The set of natural numbers is larger than the real numbers.

C) The unit interval is countable.

D) The unit interval has the power of the continuum. ✅

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether

A = [0,1]

is countable or uncountable.

Solution

The interval contains all real numbers between **0** and **1**.

Cantor proved that these numbers cannot be listed completely.

Therefore,

A is **uncountable (nondenumerable).**

Example 2

Let

A = [0,1]

and

B be a set having the same cardinality as **A**.

Determine the relationship between **A** and **B**.

Solution

Since

|A| = |B|

both sets have the same cardinality.

Therefore,

B has the power of the continuum.

Example 3

Determine whether the following statement is true.

"The unit interval has the same cardinality as the set of natural numbers."

Solution

The natural numbers are countably infinite.

The unit interval is uncountable.

Therefore,

The statement is **False**.

Example 4

Let

A = {0.1,0.2,0.3,0.4}

Determine whether **A** has the power of the continuum.

Solution

The set has only four elements.

Therefore,

It is finite.

Hence,

It does **not** have the power of the continuum.

Example 5

Suppose

A is similar to the unit interval

[0,1]

Determine the cardinality of **A**.

Solution

Sets with the same cardinality are equivalent in size.

Therefore,

A has the power of the continuum.

Example 6

Let

A = N

and

B = [0,1]

Compare their cardinalities.

Solution

The natural numbers are countably infinite.

The unit interval is uncountable.

Therefore,

|N| < |[0,1]|

Example 7

Suppose a student claims that every infinite set is countable.

Is the statement correct?

Solution

The unit interval is an infinite set.

However,

It is uncountable.

Therefore,

The statement is **False**.

Example 8

Determine whether the following set has the power of the continuum.

$\mathbf{A = \{x \in R : 0 \le x \le 1\}}$

Solution

This set is exactly the unit interval.

Therefore,

It has the **power of the continuum.**

Example 9

Suppose

A is countably infinite.

Can **A** have the same cardinality as

[0,1]?

Solution

A countably infinite set has the cardinality of natural numbers.

The unit interval is uncountable.

Therefore,

A cannot have the same cardinality as [0,1].

Example 10 (Critical VU Paper Pattern)

Let

A = [0,1]

and suppose there exists a function

f : N → A

which claims to list every element of **A**.

Determine whether such a function can be one-to-one and onto.

Solution

If such a function existed,

then the unit interval would be countable.

However,

Cantor's Diagonal Argument proves that

[0,1]

is uncountable.

Hence,

No function from **N** onto **[0,1]** can list every real number.

Therefore,

The unit interval is **nondenumerable** and has the **power of the continuum**.

LESSON 38: Real Numbers and the Power of Continuum (Part 2), Cardinal Numbers (Summarized Notes)

The **set of all real numbers** is denoted by

R

The set **R** has the same cardinality as the unit interval

I = [0,1]

Therefore,

|R| = |[0,1]| = c

where **c** is called the **power of the continuum**.

This means that the entire set of real numbers and the unit interval contain the same number of elements, even though one appears much larger.

Cardinal Number

A **cardinal number** represents the size or cardinality of a set.

Each set is assigned a symbol that represents its number of elements.

Two sets are assigned the same cardinal number if and only if they are **equipotent**, that is, if there exists a one-to-one correspondence between them.

The cardinal number of a set **A** is denoted by

|A|

or

n(A)

or

card(A)

All three notations represent the same concept.

Cardinal Numbers of Finite Sets

The empty set has cardinal number

$|\emptyset| = 0$

A set containing one element has cardinal number

1

A set containing two elements has cardinal number

2

Similarly,

The set

{1,2,...,n}

has cardinal number

n

Thus,

Finite cardinal numbers are

0,1,2,3,...

Infinite Cardinal Numbers

The cardinal numbers of infinite sets are called

Infinite Cardinal Numbers

or

Transfinite Cardinal Numbers

The cardinal number of the set of positive integers

P = {1,2,3,...}

is denoted by

$\aleph_0$ (Aleph-Null)

Therefore,

$|P| = \aleph_0$

This is the smallest infinite cardinal number.

Power of the Continuum

The cardinal number of the unit interval

I = [0,1]

is denoted by

c

This cardinal number is called the

Power of the Continuum

Since

|R| = c

the set of real numbers also has the power of the continuum.

Every interval of real numbers has the same cardinality.

For example,

(0,1)

[2,5]

(-5,8)

all have cardinality

c

Countable Sets

A set **A** is called

Denumerable or **Countably Infinite**

if

|A| = $\aleph_0$

Examples include

Natural numbers.

Positive integers.

Even positive integers.

Countable Set

A set **A** is called

Countable

if

Its cardinality is finite

or

Its cardinality is

$\aleph_0$

Thus every finite set and every denumerable set is countable.

Sets Having the Power of Continuum

A set **A** has the power of the continuum if

|A| = c

Examples include

The set of real numbers.

The unit interval.

Any interval of real numbers.

These sets are **uncountable**.

Important Exam Points

The set of real numbers is denoted by **R**.

The cardinality of the real numbers is

|R| = c

The unit interval and the real numbers have the same cardinality.

A cardinal number represents the size of a set.

The cardinality of a set is written as **|A|**, **n(A)** or **card(A)**.

Two equipotent sets have the same cardinal number.

The empty set has cardinal number **0**.

The set **{1,2,...,n}** has cardinal number **n**.

The cardinal number of positive integers is $\aleph_0$.

$\aleph_0$ is the smallest infinite cardinal number.

The power of the continuum is denoted by **c**.

Every interval of real numbers has cardinality **c**.

A denumerable set has cardinality $\aleph_0$.

A countable set is either finite or denumerable.

A set has the power of the continuum if its cardinality is **c**.

---

5 Very Important Statement-Based MCQs

1. The cardinality of the set of real numbers is

A) 0

B) $\aleph_0$

C) c ✅

D) 1

---

2. Which symbol denotes the smallest infinite cardinal number?

A) c

B) $\aleph_0$ ✅

C) n

D) ω

---

3. A set is denumerable if

A) It has cardinality c.

B) It has cardinality $\aleph_0$. ✅

C) It is finite only.

D) It has no elements.

---

4. Which of the following sets has the power of the continuum?

A) Natural numbers.

B) Integers.

C) Real numbers. ✅

D) Rational numbers.

---

5. Two sets have the same cardinal number if they are

A) Equal.

B) Finite.

C) Equipotent. ✅

D) Disjoint.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {2,4,6,8,10}

Find the cardinal number of **A**.

Solution

The set contains

5 elements.

Therefore,

|A| = 5

Example 2

Let

A = ∅

Find the cardinal number of **A**.

Solution

The empty set contains no elements.

Therefore,

|A| = 0

Example 3

Determine whether the set

N = {1,2,3,...}

is finite, countable, or has the power of the continuum.

Solution

The set of natural numbers is infinite.

Its cardinality is

$\aleph_0$

Therefore,

N is **countably infinite (denumerable).**

Example 4

Determine the cardinality of the interval

(5,10)

Solution

Every interval of real numbers has the same cardinality as

[0,1]

Therefore,

|(5,10)| = c

Example 5

Compare the cardinalities of

N

and

R

Solution

The cardinality of

N = $\aleph_0$

The cardinality of

R = c

Since

$\aleph_0$ < c

Therefore,

The set of real numbers is strictly larger than the set of natural numbers.

Example 6

Suppose

A ≈ [0,1]

Determine the cardinality of **A**.

Solution

Since **A** is equipotent to the unit interval,

both sets have the same cardinality.

Therefore,

|A| = c

Example 7

Determine whether the following statement is true.

"Every countable set has cardinality c."

Solution

A countable set has either finite cardinality or

$\aleph_0$

The cardinality

c

belongs to uncountable sets.

Therefore,

The statement is **False**.

Example 8

Let

A = (−∞,∞)

and

B = [0,1]

Compare their cardinalities.

Solution

Both sets are intervals of real numbers.

Every interval has the power of the continuum.

Therefore,

|A| = |B| = c

Example 9

Determine whether the following sets are equipotent.

A = R

B = (2,8)

Solution

Both sets have cardinality

c

Hence,

There exists a one-to-one correspondence between them.

Therefore,

A and B are equipotent.

Example 10 (Critical VU Paper Pattern)

Let

A = N

B = R

C = [0,1]

Determine the cardinality of each set and compare them.

Solution

The set

A = N

has cardinality

$|A| = \aleph_0$

The set

B = R

has cardinality

$|B| = c$

The set

C = [0,1]

also has cardinality

$|C| = c$

Since

$\aleph_0 < c$

we conclude that

|N| < |R| = |[0,1]|

Therefore,

The natural numbers are **countably infinite**, while the real numbers and the unit interval have the **power of the continuum**.

LESSON 39: Tautology, Contradiction and Logical Equivalences (Summarized Notes)

A **compound proposition** is formed by combining two or more simple propositions using logical operators.

The truth value of a compound proposition depends on the truth values of its component propositions.

Some compound propositions are always true, while others are always false.

These are called **tautologies** and **contradictions**.

Tautology

A **tautology** is a compound proposition that is **always true**, regardless of the truth values of its propositional variables.

Its final truth table contains only **True (T)** values.

One important example is

p ∨ ~p

This proposition is always true because either **p** is true or **p** is false.

Truth Table of Tautology

p	~p	p ∨ ~p
T	F	T
F	T	T

Since every entry in the final column is **True**, the proposition is a **tautology**.

Contradiction

A **contradiction** is a compound proposition that is **always false**, regardless of the truth values of its propositional variables.

Its final truth table contains only **False (F)** values.

One important example is

p ∧ ~p

A proposition and its negation cannot both be true simultaneously.

Truth Table of Contradiction

p	~p	p ∧ ~p
T	F	F
F	T	F

Since every entry in the final column is **False**, the proposition is a **contradiction**.

Logical Equivalence

Two compound propositions are **logically equivalent** if they have **exactly the same truth values** for every possible combination of truth values.

Logical equivalence is denoted by

≡

Truth tables are the standard method used to verify logical equivalence.

If the final columns of two truth tables are identical, then the propositions are logically equivalent.

De Morgan's Law

One of the most important logical equivalences is

~(p ∨ q) ≡ ~p ∧ ~q

This law states that the negation of a disjunction is equal to the conjunction of the negations.

Another important law is

~(p ∧ q) ≡ ~p ∨ ~q

These are known as **De Morgan's Laws**.

Conditional Equivalence

A conditional statement can be rewritten using disjunction.

The important equivalence is

$p \rightarrow q \equiv \sim p \lor q$

This equivalence is frequently used to simplify logical expressions and proofs.

Combining Conditional Statements

Two conditional statements having the same hypothesis can be combined.

The following logical equivalence is very important.

$(p \rightarrow q) \land (p \rightarrow r) \equiv p \rightarrow (q \land r)$

This means that if **p** implies **q** and **p** also implies **r**, then **p** implies **q and r** together.

This equivalence is commonly used in mathematical proofs.

Important Exam Points

A tautology is always true.

A contradiction is always false.

The proposition **$p \lor \sim p$** is a tautology.

The proposition **$p \land \sim p$** is a contradiction.

Truth tables are used to verify logical equivalence.

Two propositions are logically equivalent if their truth tables are identical.

Logical equivalence is denoted by ≡.

$\sim(p \lor q) \equiv \sim p \land \sim q$ is De Morgan's First Law.

$\sim(p \land q) \equiv \sim p \lor \sim q$ is De Morgan's Second Law.

$p \rightarrow q \equiv \sim p \lor q$ is the conditional equivalence.

$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ is an important logical equivalence frequently used in proofs.

---

5 Very Important Statement-Based MCQs

1. A compound proposition that is always true is called

A) Contradiction.

B) Contingency.

C) Tautology. ✅

D) Conditional statement.

---

2. Which proposition is always false?

A) $p \vee \sim p$

B) $p \wedge \sim p$ ✅

C) $p \rightarrow q$

D) $p \vee q$

---

3. Two propositions are logically equivalent when

A) They contain the same variables.

B) Their truth tables are identical. ✅

C) They have the same number of operators.

D) Both are tautologies.

---

4. Which logical equivalence is De Morgan's First Law?

A) $\sim\sim p \equiv p$

B) ~(p ∨ q) ≡ ~p ∧ ~q ✅

C) p → q ≡ ~p ∨ q

D) p ∨ p ≡ p

---

5. Which proposition is logically equivalent to p → q?

A) p ∧ q

B) ~p ∨ q ✅

C) p ∨ q

D) ~q ∨ p

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Construct the truth table for

(p ∨ ~p)

Determine whether it is a tautology or contradiction.

Solution

p	~p	p ∨ ~p
T	F	T
F	T	T

The final column contains only **True**.

Therefore,

(p ∨ ~p) is a **tautology**.

Example 2

Construct the truth table for

$(p \land \sim p)$

Determine its nature.

Solution

p	**~p**	**$p \land \sim p$**
T	F	F
F	T	F

The final column contains only **False**.

Therefore,

$(p \land \sim p)$ is a **contradiction**.

Example 3

Verify using a truth table that

$\sim(p \lor q) \equiv \sim p \land \sim q$

Solution

p	**q**	**$\sim(p \lor q)$**	**~p**	**~q**	**$\sim p \land \sim q$**
T	T	F	F	F	F
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

The last two columns are identical.

Therefore,

$\sim(p \lor q) \equiv \sim p \land \sim q$

Example 4

Verify using a truth table that

$p \rightarrow q \equiv \sim p \lor q$

Solution

p	q	p→q	~p	~p∨q
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The final columns are identical.

Therefore,

$p \rightarrow q \equiv \sim p \lor q$

Example 5

Verify whether

$(p \rightarrow q) \land (p \rightarrow r) \equiv p \rightarrow (q \land r)$

for

$p = T, q = T, r = F$

Solution

$p \rightarrow q = T$

$p \rightarrow r = F$

Therefore,

Left Side = $T \land F = F$

Also,

$q \land r = F$

Hence,

$p \rightarrow (q \land r) = T \rightarrow F = F$

Both sides are equal.

Example 6

Verify whether

$(p \to q) \land (p \to r) \equiv p \to (q \land r)$

for

p = F, q = T, r = F

Solution

$p \to q = T$

$p \to r = T$

Left Side = T

Also,

$q \land r = F$

Since

p = F,

$p \to (q \land r) = T$

Both sides are equal.

Example 7

Determine whether

$[(p \to q) \land (q \to r)] \to (p \to r)$

is true for

p = T, q = T, r = F

Solution

$p \rightarrow q = T$

$q \rightarrow r = F$

Left Side = F

$p \rightarrow r = F$

Therefore,

$F \rightarrow F = T$

Hence,

The proposition is **True** for this case.

Example 8

Using De Morgan's Law, simplify

$\sim[(p \lor q) \lor r]$

Solution

First,

$\sim[(p \lor q) \lor r]$

$= \sim(p \lor q) \land \sim r$

Applying De Morgan's Law again,

$= (\sim p \land \sim q) \land \sim r$

Therefore,

The simplified expression is

$\sim p \land \sim q \land \sim r$

Example 9

Determine the nature of

$[(p \lor \sim p) \land (q \lor \sim q)]$

Solution

Both

$p \lor \sim p$

and

$q \lor \sim q$

are tautologies.

The conjunction of two tautologies is always true.

Therefore,

The given proposition is also a **tautology**.

Example 10 (Critical VU Paper Pattern)

Construct the complete truth table for

$[(p \to q) \land (p \to r)] \leftrightarrow [p \to (q \land r)]$

Determine whether the two propositions are logically equivalent.

Solution

Evaluate all eight possible combinations of

p, q, r

Compute

$p \to q$

$p \to r$

$(p \to q) \land (p \to r)$

$q \land r$

$p \to (q \land r)$

Finally, compare both sides.

The final columns are identical for all possible truth values.

Therefore,

$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

LESSON 40: Constructing New Logical Equivalences, Converse, Contrapositive and Inverse (Summarized Notes)

Previously established logical equivalences can be used to derive **new logical equivalences**.

Instead of constructing complete truth tables every time, logical expressions can often be simplified by applying known logical laws step by step.

This method is called **constructing new logical equivalences**.

It is widely used in mathematical proofs and logical reasoning.

Constructing Logical Equivalences

Logical expressions can be simplified by repeatedly applying known equivalence laws such as

De Morgan's Laws.

Conditional Equivalence.

Double Negation Law.

Distributive Laws.

Associative Laws.

Commutative Laws.

Identity Laws.

Each step replaces an expression with another logically equivalent expression until the desired result is obtained.

Proving a Tautology Using Logical Equivalences

Instead of using a truth table, a proposition may be proved to be a **tautology** by simplifying it using logical equivalence laws.

If the final expression becomes

T

then the proposition is a tautology.

This method is usually shorter than constructing a complete truth table.

Conditional Statement

A conditional statement has the form

p → q

where

p is called the **hypothesis**.

q is called the **conclusion**.

Several important statements can be formed from a conditional statement.

These are

Converse.

Contrapositive.

Inverse.

Converse

The **converse** of

p → q

is

q → p

The hypothesis and conclusion are interchanged.

The converse is **not** logically equivalent to the original conditional statement.

Example

Original statement

If it is raining, then the home team wins.

Converse

If the home team wins, then it is raining.

Contrapositive

The **contrapositive** of

$p \rightarrow q$

is

$\sim q \rightarrow \sim p$

The hypothesis and conclusion are both negated and interchanged.

The contrapositive is **always logically equivalent** to the original conditional statement.

Example

Original statement

If it is raining, then the home team wins.

Contrapositive

If the home team does not win, then it is not raining.

Inverse

The **inverse** of

$p \rightarrow q$

is

$\sim p \rightarrow \sim q$

The hypothesis and conclusion are both negated without changing their positions.

The inverse is **not logically equivalent** to the original conditional statement.

However,

The inverse is logically equivalent to the converse.

Example

Original statement

If it is raining, then the home team wins.

Inverse

If it is not raining, then the home team does not win.

Relationships Among Conditional Statements

Original Statement

$p \rightarrow q$

Converse

$q \rightarrow p$

Inverse

$\sim p \rightarrow \sim q$

Contrapositive

$\sim q \rightarrow \sim p$

Important relationships are

$p \rightarrow q \equiv \sim q \rightarrow \sim p$

Original and contrapositive are logically equivalent.

$q \rightarrow p \equiv \sim p \rightarrow \sim q$

Converse and inverse are logically equivalent.

Important Exam Points

Logical equivalence laws can be combined to construct new equivalences.

A tautology can be proved using logical equivalence instead of a truth table.

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The converse of **p → q** is **q → p**.

The inverse of **p → q** is **~p → ~q**.

The contrapositive of **p → q** is **~q → ~p**.

The original statement and its contrapositive are logically equivalent.

The converse and inverse are logically equivalent.

The converse is generally **not** equivalent to the original statement.

The inverse is generally **not** equivalent to the original statement.

The contrapositive always has the same truth value as the original conditional statement.

---

5 Very Important Statement-Based MCQs

1. The converse of p → q is

A) ~q → ~p

B) q → p ✅

C) ~p → ~q

D) p ∧ q

---

2. The contrapositive of p → q is

A) q → p

B) ~p → ~q

C) ~q → ~p ✅

D) p ∨ q

---

3. Which statement is logically equivalent to the original conditional statement?

A) Converse.

B) Inverse.

C) Contrapositive. ✅

D) None of these.

---

4. Which two statements are logically equivalent?

A) Original statement and converse.

B) Converse and inverse. ✅

C) Original statement and inverse.

D) Converse and contrapositive.

---

5. Which method can be used to prove a tautology without constructing a truth table?

A) Mathematical induction.

B) Logical equivalence laws. ✅

C) Set notation.

D) Cartesian product.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Construct the converse, inverse, and contrapositive of

If a number is divisible by 4, then it is even.

Solution

Converse:

If a number is even, then it is divisible by 4.

Inverse:

If a number is not divisible by 4, then it is not even.

Contrapositive:

If a number is not even, then it is not divisible by 4.

Example 2

Construct the converse, inverse, and contrapositive of

If a student studies regularly, then the student passes the examination.

Solution

Converse:

If a student passes the examination, then the student studies regularly.

Inverse:

If a student does not study regularly, then the student does not pass the examination.

Contrapositive:

If a student does not pass the examination, then the student does not study regularly.

Example 3

Using logical equivalence, prove that

$(p \rightarrow q) \land p \equiv p \land q$

Solution

Replace

$p \rightarrow q$

by

$\sim p \lor q$

Therefore,

$(\sim p \lor q) \land p$

Using the distributive law,

$= (\sim p \land p) \lor (q \land p)$

Since

$\sim p \land p = F$

Therefore,

$F \lor (p \land q)$

$= p \land q$

Hence,

$(p \to q) \land p \equiv p \land q$

Example 4

Using logical equivalence, simplify

$\sim(p \to q)$

Solution

Using the conditional equivalence,

$p \to q \equiv \sim p \lor q$

Therefore,

$\sim(\sim p \lor q)$

Applying De Morgan's Law,

$= \sim\sim p \land \sim q$

Using Double Negation,

$= p \wedge \sim q$

Therefore,

$\mathbf{\sim(p \rightarrow q) \equiv p \wedge \sim q}$

Example 5

Prove that

$\mathbf{(p \wedge q) \rightarrow p}$

is a tautology using logical equivalences.

Solution

$(p \wedge q) \rightarrow p$

$\equiv \sim(p \wedge q) \vee p$

$\equiv (\sim p \vee \sim q) \vee p$

$\equiv (\sim p \vee p) \vee \sim q$

$\equiv T \vee \sim q$

$\equiv T$

Therefore,

The proposition is a **tautology**.

Example 6

Prove that

$\mathbf{p \rightarrow (p \vee q)}$

is a tautology.

Solution

$p \rightarrow (p \lor q)$

$\equiv \sim p \lor (p \lor q)$

$\equiv (\sim p \lor p) \lor q$

$\equiv T \lor q$

$\equiv T$

Therefore,

The proposition is a **tautology**.

Example 7

State the converse, inverse, and contrapositive of

A positive integer is prime only if it has no divisors other than 1 and itself.

Solution

Let

p:

A positive integer is prime.

q:

It has no divisors other than 1 and itself.

Converse:

If a positive integer has no divisors other than 1 and itself, then it is prime.

Inverse:

If a positive integer is not prime, then it has divisors other than 1 and itself.

Contrapositive:

If a positive integer has divisors other than 1 and itself, then it is not prime.

Example 8

Determine whether the converse of

If a figure is a square, then it is a rectangle

is true.

Solution

Converse:

If a figure is a rectangle, then it is a square.

A rectangle need not be a square.

Therefore,

The converse is **False**.

Example 9

Determine whether the original statement and its contrapositive are logically equivalent.

Solution

The original statement is

$p \rightarrow q$

The contrapositive is

$\sim q \rightarrow \sim p$

Both have identical truth tables.

Therefore,

They are **logically equivalent**.

Example 10 (Critical VU Paper Pattern)

Using logical equivalence laws, prove that

$[(p \rightarrow q) \land (q \rightarrow r)] \rightarrow (p \rightarrow r)$

is a tautology.

Solution

Replace each conditional statement by

$\sim p \lor q$

The proposition becomes

$[(\sim p \lor q) \land (\sim q \lor r)] \rightarrow (\sim p \lor r)$

Apply the conditional equivalence again,

De Morgan's Laws,

Distributive Laws,

Associative Laws,

and Identity Laws.

After repeated simplification,

the expression reduces to

T

Therefore,

$[(p \rightarrow q) \land (q \rightarrow r)] \rightarrow (p \rightarrow r)$

is a **tautology**.

LESSON 41: Argument, Rules of Inference for Propositional Logic and Examples (Summarized Notes)

An **argument** in propositional logic is a sequence of propositions.

All propositions except the last one are called **premises**.

The last proposition is called the **conclusion**.

The purpose of an argument is to determine whether the conclusion logically follows from the premises.

Premises and Conclusion

The statements that provide evidence are called **premises**.

The statement that follows from the premises is called the **conclusion**.

Example

Premise 1:

If you have a current password, then you can log onto the network.

Premise 2:

You have a current password.

Conclusion:

You can log onto the network.

Valid Argument

An argument is **valid** if the truth of all its premises guarantees that the conclusion is true.

A valid argument has the form

Premises

$p_1, p_2, \ldots, p_n$

Conclusion

q

The argument is valid if

$(p_1 \land p_2 \land \ldots \land p_n) \to q$

is a **tautology**.

A valid argument does **not** mean that every premise is actually true.

It only means that **if** all premises are true, then the conclusion must also be true.

Rules of Inference

Rules of inference are logical rules used to derive valid conclusions from given premises.

They form the foundation of mathematical proofs.

Modus Ponens (Law of Detachment)

Modus Ponens is one of the most important rules of inference.

Its form is

$p \to q$

p

Therefore,

q

If the conditional statement and its hypothesis are true, then the conclusion must also be true.

Example

If it snows today, then we will go skiing.

It snows today.

Therefore,

We will go skiing.

Modus Tollens

The form of Modus Tollens is

$p \to q$

~q

Therefore,

~p

If the conclusion of a conditional statement is false, then its hypothesis must also be false.

Example

If it snows today, then the university will close.

The university is not closed.

Therefore,

It did not snow today.

Addition Rule

The form is

p

Therefore,

$\mathbf{p \lor q}$

If one proposition is true, then the disjunction is also true.

Example

Salman is a mathematics major.

Therefore,

Salman is a mathematics major or a computer science major.

Simplification Rule

The form is

$\mathbf{p \land q}$

Therefore,

p

A conjunction implies each of its individual parts.

Example

Ezaan is a mathematics major and a computer science major.

Therefore,

Ezaan is a mathematics major.

Hypothetical Syllogism

The form is

$\mathbf{p \rightarrow q}$

$\mathbf{q \rightarrow r}$

Therefore,

$\mathbf{p \rightarrow r}$

This rule combines two conditional statements.

Validity of Arguments

If an argument follows a valid rule of inference, then its conclusion logically follows from its premises.

Validity depends on the logical form of the argument, not on the actual meaning of the statements.

Important Exam Points

An argument consists of premises and a conclusion.

All statements except the last are premises.

The last statement is the conclusion.

A valid argument guarantees the truth of the conclusion whenever the premises are true.

A valid argument corresponds to a tautology.

Modus Ponens has the form

$\mathbf{p \rightarrow q, p \Rightarrow q}$

Modus Tollens has the form

$p \to q, \sim q \Rightarrow \sim p$

Addition has the form

$p \Rightarrow p \lor q$

Simplification has the form

$p \land q \Rightarrow p$

Hypothetical Syllogism has the form

$p \to q, q \to r \Rightarrow p \to r$

Rules of inference are widely used in mathematical proofs.

---

5 Very Important Statement-Based MCQs

1. In an argument, the final proposition is called the

A) Premise.

B) Hypothesis.

C) Conclusion. ✅

D) Proposition.

---

2. Which rule of inference has the form $p \to q, p \Rightarrow q$?

A) Modus Tollens.

B) Addition.

C) Modus Ponens. ✅

D) Simplification.

---

3. Which rule of inference has the form $p \to q, \sim q \Rightarrow \sim p$?

A) Addition.

B) Simplification.

C) Modus Tollens. ✅

D) Resolution.

---

4. Which rule of inference allows us to conclude p ∨ q from p?

A) Addition. ✅

B) Simplification.

C) Modus Ponens.

D) Hypothetical Syllogism.

---

5. An argument is valid if

A) All premises are false.

B) The conclusion is always false.

C) The premises logically imply the conclusion. ✅

D) The conclusion is a contradiction.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Determine whether the following argument is valid.

Premise 1:

If a student studies regularly, then the student passes the examination.

Premise 2:

The student studies regularly.

Conclusion:

The student passes the examination.

Solution

Let

p:

The student studies regularly.

q:

The student passes the examination.

The argument has the form

$\mathbf{p \rightarrow q}$

p

Therefore,

q

This is **Modus Ponens**.

Hence,

The argument is **valid**.

Example 2

Determine whether the following argument is valid.

Premise 1:

If a number is divisible by 6, then it is divisible by 3.

Premise 2:

The number is not divisible by 3.

Conclusion:

The number is not divisible by 6.

Solution

Let

p:

The number is divisible by 6.

q:

The number is divisible by 3.

The argument has the form

p → q

~q

Therefore,

~p

This is **Modus Tollens**.

Hence,

The argument is **valid**.

Example 3

Identify the rule of inference.

Premise:

Ali is an engineer.

Conclusion:

Ali is an engineer or a teacher.

Solution

The argument has the form

p

Therefore,

p ∨ q

This is the **Addition Rule**.

Example 4

Identify the rule of inference.

Premise:

Sara is intelligent and hardworking.

Conclusion:

Sara is intelligent.

Solution

The argument has the form

p ∧ q

Therefore,

p

This is the **Simplification Rule**.

Example 5

Determine the validity of the following argument.

Premise 1:

If a function is differentiable, then it is continuous.

Premise 2:

If a function is continuous, then it is integrable on [a,b].

Conclusion:

If a function is differentiable, then it is integrable on [a,b].

Solution

The argument has the form

$p \to q$

$q \to r$

Therefore,

$p \to r$

This is **Hypothetical Syllogism**.

Hence,

The argument is **valid**.

Example 6

Find the argument form.

Premise 1:

If Socrates is human, then Socrates is mortal.

Premise 2:

Socrates is human.

Conclusion:

Socrates is mortal.

Solution

Let

p:

Socrates is human.

q:

Socrates is mortal.

The argument form is

$\mathbf{p \rightarrow q}$

p

Therefore,

q

This is **Modus Ponens**.

Hence,

The argument is **valid**.

Example 7

Use rules of inference to derive the conclusion.

Premise 1:

If Khan works hard, then he is a dull boy.

Premise 2:

If Khan is a dull boy, then he will not get a job.

Premise 3:

Khan works hard.

Conclusion:

Khan will not get a job.

Solution

From

Premise 1

and

Premise 3,

Using **Modus Ponens**,

Khan is a dull boy.

Using

Premise 2

and the above result,

Again applying **Modus Ponens**,

Khan will not get a job.

Hence,

The conclusion follows logically.

Example 8

Determine the rule of inference.

Premise 1:

If a graph is complete, then every pair of vertices is adjacent.

Premise 2:

The graph is complete.

Conclusion:

Every pair of vertices is adjacent.

Solution

The argument has the form

p → q

p

Therefore,

q

This is **Modus Ponens**.

Example 9

Determine whether the conclusion follows.

Premise 1:

If a matrix is invertible, then its determinant is non-zero.

Premise 2:

The determinant of the matrix is zero.

Conclusion:

The matrix is not invertible.

Solution

The argument has the form

$p \to q$

$\sim q$

Therefore,

$\sim p$

This is **Modus Tollens**.

Hence,

The argument is **valid**.

Example 10 (Critical VU Paper Pattern)

Given the premises

If a program compiles successfully, then it is free of syntax errors.

If a program is free of syntax errors, then it can be executed.

The program compiles successfully.

Use rules of inference to prove the conclusion

The program can be executed.

Solution

Let

p:

The program compiles successfully.

q:

The program is free of syntax errors.

r:

The program can be executed.

From

$\mathbf{p \rightarrow q}$

and

$\mathbf{q \rightarrow r}$

Using **Hypothetical Syllogism**,

We obtain

$\mathbf{p \rightarrow r}$

Since

$\mathbf{p}$

is true,

Applying **Modus Ponens**,

We conclude

r

Therefore,

The program can be executed.

LESSON 42: Propositional Functions, Quantifiers and Universal Quantifier (Summarized Notes)

A **propositional function** (also called an **open sentence**) is an expression containing one or more variables.

It does **not** have a truth value until values are assigned to its variables.

When a value from the domain is substituted for the variable, the propositional function becomes a **proposition**, which is either **True** or **False**.

Propositional Function

Let **A** be a given set.

A propositional function is written as

p(x)

where **x ∈ A**.

For every element of **A**, the expression **p(x)** becomes a statement with a definite truth value.

The set **A** is called the **domain** of the propositional function.

Example

Let

A = {1,2,3,4,5}

and

p(x): x > 3

Then

p(2) = False

p(4) = True

p(5) = True

Domain

The **domain** is the set of all possible values that can be substituted into the variable.

Every value assigned to the variable must belong to the domain.

Truth Set

The **truth set** of a propositional function is the set of all elements in the domain for which the propositional function is **True**.

If

T = {x ∈ A : p(x) is True}

then **T** is called the **truth set** of **p(x)**.

Example

Let

A = {1,2,3,4,5,6}

and

p(x): x is even

Then

Truth Set

T = {2,4,6}

Propositional Functions with Two Variables

A propositional function may contain more than one variable.

It is written as

p(x,y)

A truth value is obtained only after assigning values to **both variables**.

Example

p(x,y): x + y = 8

If

x = 3

y = 5

Then

p(3,5) = True

If

x = 2

y = 3

Then

p(2,3) = False

Quantifiers

A **quantifier** specifies how many elements in the domain satisfy a propositional function.

The two most important quantifiers are

Universal Quantifier.

Existential Quantifier.

Universal Quantifier

The **Universal Quantifier** is denoted by

∀

It means

"For all" or **"For every."**

If

$\forall x\, p(x)$

is written,

it means

p(x) is true for every element of the domain.

A universal statement is **True** only if every element satisfies the propositional function.

If even one element does not satisfy the condition,

the universal statement is **False**.

Example

Let

A = {2,4,6,8}

and

p(x): x is even

Then

$\forall x\, p(x)$

is **True**.

Important Exam Points

A propositional function is also called an **open sentence**.

A propositional function has no truth value until values are assigned.

The domain is the set of allowable values.

The truth set contains all elements for which the propositional function is true.

A propositional function with one variable is written as **p(x)**.

A propositional function with two variables is written as **p(x,y)**.

The universal quantifier is denoted by ∀.

∀ means **"for every"** or **"for all."**

A universal statement is true only if every element satisfies the condition.

If one element violates the condition, the universal statement becomes false.

---

5 Very Important Statement-Based MCQs

1. A propositional function is also called an

A) Argument.

B) Open sentence. ✅

C) Tautology.

D) Contradiction.

---

2. The set of all allowable values of a variable is called the

A) Truth set.

B) Range.

C) Domain. ✅

D) Codomain.

---

3. The truth set of p(x) contains

A) All elements of the domain.

B) Only those elements for which **p(x)** is true. ✅

C) Only false statements.

D) Only finite elements.

---

4. The symbol ∀ represents

A) There exists.

B) For all. ✅

C) Therefore.

D) Logical equivalence.

---

5. A universal statement is false when

A) Every element satisfies the condition.

B) At least one element does not satisfy the condition. ✅

C) The domain is finite.

D) The truth set is empty.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {1,2,3,4,5,6}

Define

p(x): x > 4

Find the truth value of

p(3)

and

p(6)

Solution

p(3):

3 > 4

False

p(6):

6 > 4

True

Example 2

Let

A = {1,2,3,4,5,6,7,8}

Define

p(x): x is divisible by 3

Find the truth set.

Solution

The elements divisible by 3 are

3

6

Therefore,

Truth Set

{3,6}

Example 3

Let

A = {−3,−2,−1,0,1,2,3}

Define

p(x): $x^2 \leq 4$

Find the truth set.

Solution

Evaluate each element.

$(-3)^2 = 9$ ✗

$(-2)^2 = 4$ ✓

$(-1)^2 = 1$ ✓

$0^2 = 0$ ✓

$1^2 = 1$ ✓

$2^2 = 4$ ✓

$3^2 = 9$ ✗

Therefore,

Truth Set

{−2,−1,0,1,2}

Example 4

Let

A = {1,2,3,4,5}

Determine whether

$\forall x(x < 6)$

is true.

Solution

Every element is less than 6.

Therefore,

The statement is **True**.

Example 5

Let

A = {2,4,6,7,8}

Determine whether

∀x(x is even)

is true.

Solution

The element

7

is not even.

Therefore,

The statement is **False**.

Example 6

Let

p(x,y): $x^2 + y^2 = 25$

Determine the truth values of

p(3,4)

and

p(4,4)

Solution

p(3,4):

9 + 16 = 25

True

p(4,4):

16 + 16 = 32

False

Example 7

Let

A = {−2,−1,0,1,2}

Define

p(x): $x^3 \geq 0$

Find the truth set.

Solution

Evaluate each element.

$(-2)^3 = -8$ ✗

$(-1)^3 = -1$ ✗

$0^3 = 0$ ✓

$1^3 = 1$ ✓

$2^3 = 8$ ✓

Therefore,

Truth Set

{0,1,2}

Example 8

Let

A = {1,2,3,4,5,6}

Define

p(x): x is prime

Find the truth set.

Solution

Prime numbers are

2

3

5

Therefore,

Truth Set

{2,3,5}

Example 9

Let

p(x,y): x divides y

Determine the truth values of

p(4,20)

and

p(6,20)

Solution

4 divides 20.

Therefore,

p(4,20) = True

6 does not divide 20.

Therefore,

p(6,20) = False

Example 10 (Critical VU Paper Pattern)

Let

A = {−3,−2,−1,0,1,2,3}

Define

p(x): $x^2 + x \geq 0$

Find the truth set and determine whether

$\forall x\ p(x)$

is true.

Solution

Evaluate each element.

x = -3

9 - 3 = 6 ✓

x = -2

4 - 2 = 2 ✓

x = -1

1 - 1 = 0 ✓

x = 0

0 ✓

x = 1

2 ✓

x = 2

6 ✓

x = 3

12 ✓

Truth Set

{−3,−2,−1,0,1,2,3}

Since every element belongs to the truth set,

∀ x p(x)

is **True**.

LESSON 43: Existential Quantifier and Negation of Quantified Statements (Summarized Notes)

A **quantifier** tells us how many elements of the domain satisfy a propositional function.

The two main quantifiers are:

Universal Quantifier (∀) – "For all"

Existential Quantifier (∃) – "There exists"

This lesson focuses on the **Existential Quantifier** and the **Negation of Quantified Statements**.

Existential Quantifier

The **Existential Quantifier** is denoted by

∃

It means

"There exists" or **"There is at least one."**

If

$\exists$ x p(x)

is written,

it means

There is at least one element in the domain for which p(x) is true.

Unlike the universal quantifier, the existential quantifier requires **only one** element to satisfy the condition.

Example

Let

A = {1,2,3,4,5}

and

p(x): x is even

Since

2 and 4 are even,

there exists at least one element satisfying the condition.

Therefore,

$\exists$ x p(x)

is **True**.

Truth Value of an Existential Statement

An existential statement is **True** if **at least one** element in the domain satisfies the propositional function.

It is **False** only if **no element** satisfies the condition.

Example

Let

A = {1,3,5,7}

and

p(x): x is even

None of the elements are even.

Therefore,

∃ x p(x)

is **False**.

Negation of Quantified Statements

The **negation** of a quantified statement changes both the **quantifier** and the **statement**.

The negation of a universal statement becomes an existential statement.

The negation of an existential statement becomes a universal statement.

Negation of Universal Quantifier

If

∀ x p(x)

is given,

then its negation is

∃ x ¬p(x)

which means

There exists at least one element for which p(x) is false.

Example

Statement

$\forall x(x > 0)$

Negation

$\exists x(x \leq 0)$

Meaning

There exists at least one number that is not greater than zero.

Negation of Existential Quantifier

If

$\exists x\, p(x)$

is given,

then its negation is

$\forall x\, \neg p(x)$

which means

For every element, p(x) is false.

Example

Statement

$\exists x(x < 0)$

Negation

$\forall x(x \geq 0)$

Meaning

Every element is greater than or equal to zero.

Rules for Negation

Negation changes

$\forall \rightarrow \exists$

$\exists \rightarrow \forall$

and also changes the predicate to its opposite.

Examples

$x > 5$ becomes $x \leq 5$

$x \geq 4$ becomes $x < 4$

$x = 3$ becomes $x \neq 3$

$x \neq 7$ becomes $x = 7$

Difference Between Universal and Existential Quantifiers

Universal Quantifier (∀)	Existential Quantifier (∃)
Means "For all"	Means "There exists"
Every element must satisfy the condition	At least one element must satisfy the condition
False if one element fails	True if one element satisfies

Important Exam Points

The existential quantifier is denoted by ∃ .

∃ means **"There exists"** or **"At least one."**

An existential statement is true if at least one element satisfies the condition.

An existential statement is false only when no element satisfies the condition.

The negation of ∀ becomes ∃ .

The negation of ∃ becomes ∀.

Negation changes both the quantifier and the predicate.

Remember

$\neg(\forall x\, p(x)) = \exists x\, \neg p(x)$

$\neg(\exists x\, p(x)) = \forall x\, \neg p(x)$

These two formulas are among the most frequently tested concepts in **Virtual University Midterm and Final examinations**.

---

5 Very Important Statement-Based MCQs

1. The symbol ∃ represents

A) For all

B) Therefore

C) There exists. ✅

D) Logical equivalence

---

2. An existential statement is true if

A) Every element satisfies the condition.

B) At least one element satisfies the condition. ✅

C) The domain is finite.

D) The truth set is empty.

---

3. The negation of $\forall x\, p(x)$ is

A) $\forall x\, \neg p(x)$

B) $\exists x\, \neg p(x)$ ✅

C) ∃ x p(x)

D) None of these

---

4. The negation of ∃ x p(x) is

A) ∀ x ¬p(x) ✅

B) ∃ x ¬p(x)

C) ∀ x p(x)

D) None of these

---

5. An existential statement becomes false when

A) At least one element satisfies the condition.

B) Every element satisfies the condition.

C) No element satisfies the condition. ✅

D) The domain has one element.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Let

A = {1,2,3,4,5}

Define

p(x): x > 4

Determine whether

∃ x p(x)

is true.

Solution

Only

5 > 4

Therefore,

∃ x p(x) is **True**.

Example 2

Let

A = {2,4,6,8}

Define

p(x): x is odd

Determine the truth value.

Solution

No element is odd.

Therefore,

∃ x p(x) is **False**.

Example 3

Let

A = {−2,−1,0,1,2}

Define

p(x): x < 0

Determine whether

$\exists x\, p(x)$

is true.

Solution

−2 and −1 satisfy the condition.

Therefore,

True.

Example 4

Negate

$\forall x(x > 5)$

Solution

Negation:

$\exists x(x \leq 5)$

Example 5

Negate

$\exists x(x^2 = 9)$

Solution

Negation:

$\forall x(x^2 \neq 9)$

Example 6

Let

A = {3,6,9,12}

Determine whether

∃ x(x is divisible by 5)

is true.

Solution

None of the elements are divisible by 5.

Therefore,

False.

Example 7

Negate

$\forall x(x \neq 0)$

Solution

Negation:

$\exists x(x = 0)$

Example 8

Negate

$\exists x(x \leq 10)$

Solution

Negation:

$\forall x(x > 10)$

Example 9

Let

A = {1,2,3,4,5}

Determine whether

∃ $x(x^2 = 16)$

is true.

Solution

$4^2 = 16$

Therefore,

True.

Example 10 (Critical VU Paper Pattern)

Let

A = {−3,−2,−1,0,1,2,3}

Define

$p(x)$: $x^2 < 0$

Determine

Whether **∃ x p(x)** is true.

The negation of the statement.

Solution

For every real number,

$x^2 \geq 0$

Hence,

No element satisfies

$x^2 < 0$

Therefore,

$\exists x\, p(x)$ is **False**.

Negation:

$\forall x(x^2 \geq 0)$

This statement is **True**.

LESSON 44: Introduction to Proofs and Methods of Proving Theorems (Part 1) (Summarized Notes)

A **proof** is a logical argument that demonstrates a mathematical statement is **True**.

Proofs are based on **definitions, axioms, previously proved theorems, and logical reasoning**.

The purpose of a proof is to establish the validity of a mathematical statement beyond any doubt.

What is a Theorem?

A **theorem** is a mathematical statement that has been proved to be true.

Before a theorem is accepted, it must be supported by a valid mathematical proof.

Related Mathematical Terms

Axiom (Postulate)

An **axiom** is a statement accepted as true **without proof**.

It serves as the starting point for mathematical reasoning.

Example

Through any two distinct points, there is exactly one straight line.

Definition

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A **definition** explains the meaning of a mathematical concept.

Definitions are used in constructing mathematical proofs.

Example

An **even integer** is an integer of the form

2k

where **k** is an integer.

Lemma

A **lemma** is a theorem proved mainly to help prove another theorem.

It is often called a **helper theorem**.

Corollary

A **corollary** is a statement that follows directly from an already proved theorem.

It usually requires little or no additional proof.

What is a Proof?

A proof is a sequence of logical steps that begins with known facts and ends by establishing the truth of the required statement.

A valid proof must contain:

Clearly stated assumptions.

Logical reasoning.

Correct mathematical steps.

A justified conclusion.

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Methods of Proving Theorems

Several proof techniques are used in mathematics.

The most common methods are:

Direct Proof.

Proof by Contrapositive.

Proof by Contradiction.

Proof by Cases.

Mathematical Induction.

Part 1 mainly introduces the **Direct Proof Method**.

Direct Proof

A **direct proof** starts with the given assumptions (hypotheses) and uses logical reasoning to reach the required conclusion.

If the statement is of the form

$\mathbf{p \rightarrow q}$

we assume **p** is true and prove **q**.

This is the simplest and most commonly used proof method.

Steps of Direct Proof

Assume the given hypothesis is true.

Apply definitions and known theorems.

Perform logical mathematical steps.

Reach the desired conclusion.

Example 1

Prove that

The sum of two even integers is even.

Proof

Let

a = 2m

b = 2n

where

m and n are integers.

Then

a + b

= 2m + 2n

= 2(m + n)

Since

m + n

is an integer,

a + b

is divisible by 2.

Therefore,

the sum of two even integers is **even**.

Example 2

Prove that

The sum of two odd integers is even.

Proof

Let

$a = 2m + 1$

$b = 2n + 1$

Then

$a + b$

$= (2m + 1) + (2n + 1)$

$= 2m + 2n + 2$

$= 2(m + n + 1)$

Since

$m + n + 1$

is an integer,

the sum is **even**.

Example 3

Prove that

If x is an even integer, then x^2 is even.

Proof

Let

$x = 2k$

Then

x^2

$= (2k)^2$

$= 4k^2$

$= 2(2k^2)$

Therefore,

x^2 is **even**.

Structure of a Mathematical Proof

Every proof generally consists of:

Given

The information provided.

To Prove

The statement that must be established.

Proof

The logical argument.

Conclusion

The final verified statement.

Important Exam Points

A proof is a logical argument showing that a statement is true.

A theorem is a statement proved using mathematical reasoning.

An axiom is accepted as true without proof.

A definition explains the meaning of a mathematical term.

A lemma helps prove another theorem.

A corollary follows directly from a theorem.

A direct proof begins with the hypothesis.

In a direct proof of

p → q

assume **p** and prove **q**.

Every proof must consist of logical and justified steps.

---

5 Very Important Statement-Based MCQs

1. A theorem is

A) A statement accepted without proof.

B) A statement proved using logical reasoning. ✅

C) A mathematical symbol.

D) A conjecture.

---

2. An axiom is

A) A proved theorem.

B) A statement accepted without proof. ✅

C) A contradiction.

D) A lemma.

---

3. A direct proof begins by assuming

A) The conclusion.

B) The negation of the conclusion.

C) The hypothesis is true. ✅

D) The theorem is false.

---

4. A corollary is

A) A mathematical definition.

B) A theorem used to prove another theorem.

C) A result that follows directly from a theorem. ✅

D) A contradiction.

---

5. A lemma is

A) A helper theorem used in proving another theorem. ✅

B) A false statement.

C) A proposition with no proof.

D) A logical operator.

---

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Prove that the sum of two multiples of 3 is divisible by 3.

Solution

Let

a = 3m

b = 3n

Then

a + b

= 3m + 3n

= 3(m + n)

Since

$m + n$

is an integer,

the sum is divisible by 3.

Example 2

Prove that the product of two even integers is even.

Solution

Let

$a = 2m$

$b = 2n$

Then

ab

$= (2m)(2n)$

$= 4mn$

$= 2(2mn)$

Hence,

the product is even.

Example 3

Prove that the square of an odd integer is odd.

Solution

Let

$x = 2k + 1$

Then

x^2

$= (2k + 1)^2$

$= 4k^2 + 4k + 1$

$= 2(2k^2 + 2k) + 1$

Hence,

x^2 is odd.

Example 4

Prove that the sum of an even and an odd integer is odd.

Solution

Let

$a = 2m$

$b = 2n + 1$

Then

$a + b$

$= 2m + 2n + 1$

$= 2(m + n) + 1$

Hence,

the sum is odd.

Example 5

Prove that if x is divisible by 5, then 2x is divisible by 5.

Solution

Let

$x = 5k$

Then

$2x = 10k = 5(2k)$

Hence,

2x is divisible by 5.

Example 6

Prove that the product of two odd integers is odd.

Solution

Let

$a = 2m + 1$

$b = 2n + 1$

Then

ab

$= (2m + 1)(2n + 1)$

$= 4mn + 2m + 2n + 1$

$= 2(2mn + m + n) + 1$

Hence,

the product is odd.

Example 7

Prove that the sum of three even integers is even.

Solution

Let

a = 2m,

b = 2n,

c = 2p

Then

a + b + c

= 2(m + n + p)

Hence,

the sum is even.

Example 8

Prove that if x is divisible by 4, then x is even.

Solution

Let

x = 4k

= 2(2k)

Hence,

x is even.

Example 9

Prove that the sum of two consecutive integers is odd.

Solution

Let

the integers be

n

and

n + 1

Then

n + (n + 1)

= 2n + 1

Hence,

the sum is odd.

Example 10 (Critical VU Paper Pattern)

Prove that the sum of two rational numbers is rational.

Solution

Let

a = p/q

and

b = r/s

where

q ≠ 0,

s ≠ 0,

and

p, q, r, s

are integers.

Then

a + b

= (ps + rq)/qs

Since

$ps + rq$

and

qs

are integers and

$qs \neq 0$,

the result is a rational number.

Therefore,

the sum of two rational numbers is **rational**.

LESSON 45: Methods of Proving Theorems (Part 2) (Summarized Notes)

In **Lesson 44**, we studied the **Direct Proof Method**.

In this lesson, we study two more important proof techniques:

Proof by Contrapositive

Proof by Contradiction

These methods are especially useful when a direct proof is difficult.

Proof by Contrapositive

A **contrapositive proof** proves a conditional statement by proving its **contrapositive** instead.

If the statement is

$\mathbf{p \rightarrow q}$

its contrapositive is

$\mathbf{\neg q \rightarrow \neg p}$

A statement and its contrapositive are **logically equivalent**, meaning they always have the same truth value.

Therefore, proving the contrapositive also proves the original statement.

Steps of Proof by Contrapositive

Write the original statement.

Form its contrapositive.

Assume **¬q** is true.

Use logical reasoning to prove **¬p**.

Conclude that the original statement is true.

Example 1

Prove that

If an integer is odd, then its square is odd.

Proof by Contrapositive

Contrapositive:

If **x^2 is even**, then **x is even**.

Let

x^2

be even.

Suppose

x

were odd.

Then

$x = 2k + 1$

Therefore,

x^2

$= (2k + 1)^2$

$= 4k^2 + 4k + 1$

$= 2(2k^2 + 2k) + 1$

which is odd.

This contradicts the assumption that

x^2

is even.

Hence,

x

must be even.

Therefore,

the original statement is proved.

Proof by Contradiction

A **proof by contradiction** assumes that the statement to be proved is **false**.

If this assumption leads to a contradiction, then the original statement must be true.

Steps of Proof by Contradiction

Assume the statement is false.

Apply logical reasoning.

Reach a contradiction.

Reject the false assumption.

Conclude that the original statement is true.

Example 2

Prove that

$\sqrt{2}$ is irrational.

Proof

Assume the opposite.

Suppose

$\sqrt{2}$

is rational.

Then

$\sqrt{2} = a/b$

where

a and b

have no common factor.

Squaring both sides,

$2 = a^2/b^2$

Therefore,

$a^2 = 2b^2$

Hence,

a^2

is even.

Therefore,

a

is even.

Let

$a = 2k$

Substitute,

$4k^2 = 2b^2$

or

$b^2 = 2k^2$

Hence,

b

is also even.

Thus,

both

a

and

b

are even.

This contradicts the assumption that

a/b

is in lowest terms.

Therefore,

$\sqrt{2}$

is irrational.

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Contrapositive vs Contradiction

Proof by Contrapositive	Proof by Contradiction
Proves **¬q → ¬p**	Assumes the statement is false
Uses logical equivalence	Uses contradiction
No contradiction is required	Ends with a contradiction
Often easier than direct proof	Useful for difficult theorems

Logical Equivalence

The statements

p → q

and

¬q → ¬p

are logically equivalent.

Therefore,

proving either one proves the other.

When to Use Each Method

Direct Proof

Use when the conclusion follows naturally from the hypothesis.

Contrapositive

Use when proving the opposite conclusion is easier.

Contradiction

Use when assuming the opposite quickly produces an impossible result.

Important Exam Points

The contrapositive of

$p \to q$

is

$\neg q \to \neg p$.

A statement and its contrapositive are logically equivalent.

Proof by contrapositive proves

$\neg q \to \neg p$.

Proof by contradiction begins by assuming the statement is false.

A contradiction proves the original statement is true.

The irrationality of

$\sqrt{2}$

is one of the most common examples of proof by contradiction.

---

5 Very Important Statement-Based MCQs

1. The contrapositive of

$p \to q$

is

A) $q \to p$

B) $\neg p \to \neg q$

C) $\neg q \to \neg p$ ✅

D) $q \to \neg p$

---

2. Proof by contradiction starts by assuming

A) The hypothesis is true.

B) The conclusion is true.

C) The statement is false. ✅

D) The theorem is already proved.

---

3. A contradiction in a proof means

A) The theorem is false.

B) The assumption is incorrect. ✅

C) The proof is incomplete.

D) The hypothesis is false.

---

4. Proof by contrapositive proves

A) $p \to q$

B) $q \to p$

C) $\neg q \to \neg p$ ✅

D) $\neg p \to q$

---

5. Which proof method is commonly used to prove that

$\sqrt{2}$

is irrational?

A) Direct Proof

B) Mathematical Induction

C) Proof by Contradiction ✅

D) Proof by Cases

MTH104 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

10 Very Important Mathematical Examples (University-Level / VU Paper Pattern)

Example 1

Write the contrapositive of

"If a number is divisible by 4, then it is even."

Solution

Contrapositive:

"If a number is not even, then it is not divisible by 4."

Example 2

Write the contrapositive of

"If $x > 5$, then $x > 2$."

Solution

Contrapositive:

"If $x \leq 2$, then $x \leq 5$."

Example 3

Prove:

"If n^2 is even, then n is even."

Solution

Use proof by contrapositive.

Assume

n

is odd.

Then

$n = 2k + 1$

Therefore,

$n^2 = 2(2k^2 + 2k) + 1$

which is odd.

Hence,

if

n^2

is even,

n

must be even.

Example 4

Write the contrapositive of

"If a triangle is equilateral, then it is isosceles."

Solution

"If a triangle is not isosceles, then it is not equilateral."

Example 5

Prove that there is no smallest positive rational number.

Solution

Assume

r

is the smallest positive rational number.

Then

r/2

is also positive and rational.

Since

$r/2 < r$,

this contradicts the assumption.

Therefore,

no smallest positive rational number exists.

Example 6

Prove that

0.999...

= 1

by contradiction.

Solution

Assume

0.999...

$\neq 1$.

Subtracting gives a positive difference.

However,

the difference equals

0,

which is impossible.

Therefore,

0.999...

= 1.

Example 7

Write the contrapositive of

"If $x = 0$, then $x^2 = 0$."

Solution

"If $x^2 \neq 0$, then $x \neq 0$."

Example 8

Prove that

"If an integer is divisible by 6, then it is divisible by 3."

Solution

Let

$n = 6k$

Then

$n = 3(2k)$

Hence,

n

is divisible by

1.

(Direct proof; included for comparison.)

Example 9

Prove that

"There is no greatest integer."

Solution

Assume

N

is the greatest integer.

Then

N + 1

is also an integer.

Since

N + 1 > N,

this contradicts the assumption.

Therefore,

there is no greatest integer.

Example 10 (Critical VU Paper Pattern)

Prove that

"If an integer is not divisible by 2, then it is odd."

Solution

Assume

an integer

n

is not divisible by

By the Division Algorithm,

every integer is either

2k

or

2k + 1.

Since

n

is not divisible by

2,

it cannot be

2k.

Therefore,

n = 2k + 1,

which is the definition of an odd integer.

Hence,

the statement is proved.

THE END

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